

# K-FIELD DUAL-AXIS KINETIC MASS ANALYSIS

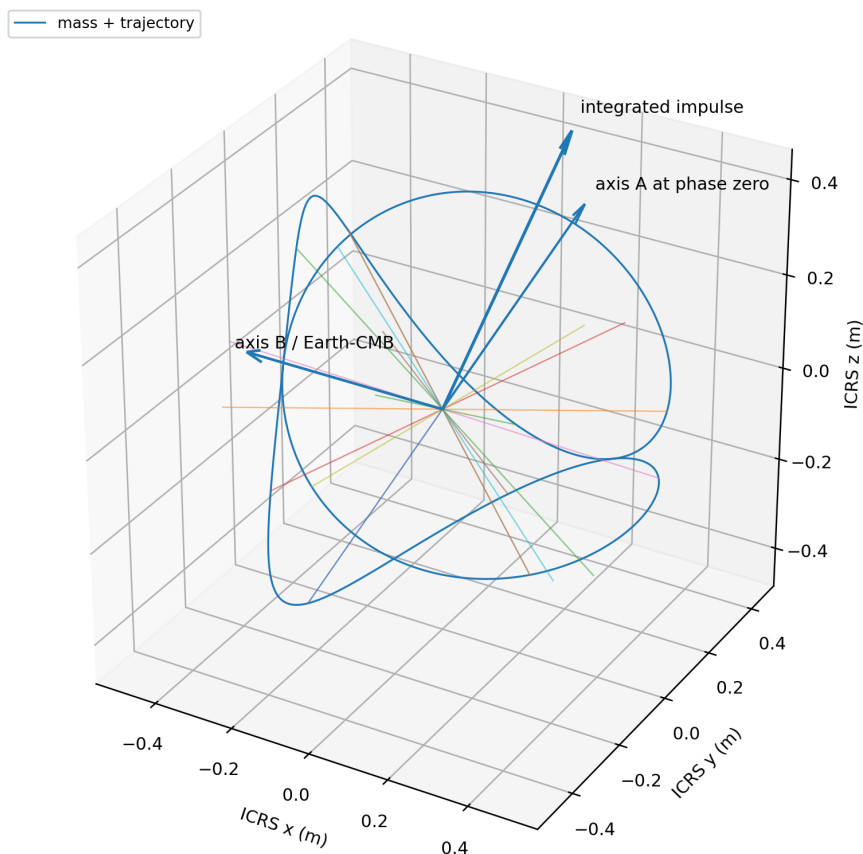
Exact 2:1 compound-rotation geometry, Q-metric kinetic energy, individual force vectors, system net force, and closed-cycle impulse in the Earth-CMB frame

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Exact 2:1 dual-axis trajectory and closed-cycle impulse direction



Title figure. Dimensionally correct dual-axis trajectory for two 1 kg point masses at radius 0.5 m. Axis B is locked to the Earth-CMB direction; axis A is perpendicular and rotates with the assembly. The displayed impulse direction is computed from the complete force integral over one axis-B rotation.

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## Abstract

This reference develops the complete mechanics of a dual-axis rotating mass pair in the K-Field quadratic phase-transition metric. Two 1 kg point masses occupy the ends of a 1 m massless bar. The bar center is fixed. Axis B is aligned with the Earth-CMB velocity vector, axis A is perpendicular to axis B, and the inner rotation rate is exactly twice the outer rotation rate. The resulting endpoint path is a closed spherical curve containing first-, second-, and third-harmonic spatial components.

The laboratory is not stationary in the K-Field frame. Each endpoint velocity is the vector sum of the common Earth-CMB translation and an equal, opposite rotational tangent velocity. The directional kinetic energy is evaluated with the authoritative positive-definite matrix  $Q$ . Its translation-rotation cross term changes sign between the two masses and creates unequal  $Q$ -metric kinetic energies. The radial derivative of that energy gives the individual force vectors. When the two forces are added, the equal pure-rotation terms cancel and the Earth-translation cross terms reinforce.

The complete force time series is evaluated at one-degree increments for one outer rotation. Under the declared unit-rate normalization  $\omega_B=1$  rad/s and  $\kappa_F=1$ , the maximum net-force magnitude is  $4.144924311 \times 10^6$  N. The force integral over the cycle is obtained both numerically and analytically. The exact impulse is  $\pi \kappa_F m$  times the cross product of the Earth-CMB axis with  $Q V_E$ . Its vector is (118842.532043, 1146514.977540, 1012275.514802) N s, its magnitude is  $1.534054060 \times 10^6$  N s, and it is exactly transverse to the Earth-CMB axis.

**Central result. The 2:1 compound rotation converts the  $Q$ -metric Earth-velocity cross term into a nonzero transverse closed-cycle impulse. The impulse is independent of rotation speed and mass radius for the defined radial force law; its cycle-average force is proportional to rotation speed.**

**Keywords:** K-Field, Cell3, Earth-CMB velocity, quadratic phase metric, dual-axis rotation, kinetic-energy anisotropy, radial force, net impulse.

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## 1. Purpose, apparatus, and scope

The analysis concerns a rigid geometric constraint rather than a freely moving particle. A massless bar of total length 1 m carries two identical 1 kg point masses at its endpoints. The center of the bar is fixed at the intersection of two rotation axes. Each mass therefore remains at radius  $R=0.5$  m from the center. The assembly has two simultaneous rotations.

| Quantity                      | Definition                                | Value                            |
|-------------------------------|-------------------------------------------|----------------------------------|
| Masses                        | Two equal point masses                    | 1 kg each                        |
| Bar                           | Rigid and massless                        | 1 m total length                 |
| Mass radius                   | Half the bar length                       | 0.5 m                            |
| Axis B                        | Outer axis, locked to Earth-CMB direction | angle to Earth-CMB = 0 deg       |
| Axis A                        | Inner axis through the center             | perpendicular to axis B          |
| Rate lock                     | Inner to outer angular-speed ratio        | $\omega_A / \omega_B = 2$        |
| Numerical force normalization | Unit outer rate and unit coupling         | $\omega_B=1$ rad/s; $\kappa_F=1$ |

The absolute rotation speed and the final engineering calibration gain are explicit scaling variables. The time-series values in this report use  $\omega_B=1$  rad/s and  $\kappa_F=1$ . This is a complete Q-metric mechanical normalization: Q is dimensionless,  $m v^2/Q$  has energy units, and its radial derivative has force units. Any calibrated implementation multiplies the tabulated forces and impulses by its measured  $\kappa_F$ .

## 2. ICRS coordinate lock and Earth-CMB motion

All vectors are expressed in the fixed ICRS Cartesian frame. Axis B is the unit Earth-CMB traversal vector. The Earth-CMB velocity is the exact-SI rendering used in the source derivation. Its direction is the same as axis B; only its scalar magnitude distinguishes the velocity vector from the unit direction.

| Object                   | ICRS x          | ICRS y         | ICRS z          | Magnitude    |
|--------------------------|-----------------|----------------|-----------------|--------------|
| Earth-CMB unit axis      | -0.970770855836 | 0.207348009111 | -0.120874929482 | 1            |
| Earth-CMB velocity (m/s) | -3.590032e+05   | 76679.886034   | -44701.060101   | 3.698125e+05 |
| Bar reference $e_0$      | -0.20086488411  | -0.42623147995 | 0.88203175897   | 1            |
| Axis A reference         | 0.131366829113  | 0.880530254232 | 0.455422032396  | 1            |

The triad ( $e_0$ ,  $A_0$ ,  $c_{\hat{}}$ ) is orthonormal and right-handed. Axis A at outer phase zero is  $A_0=c_{\hat{}} \times e_0$ . As the assembly rotates about axis B, the entire inner rotation plane is carried around the Earth-CMB axis.

## 3. Authoritative Cell3 base cell and plane system

The quadratic operator and the Earth-CMB direction are embedded in the same ICRS frame as the Cell3 base geometry. The Cell3 basis vectors are the three columns a, b, and c below. Their centered half-step combinations generate the 27-node registry. The eight combinations with every coefficient equal to plus or minus one-half are the boundary vertices.

| Basis vector | ICRS x          | ICRS y          | ICRS z          | Length         |
|--------------|-----------------|-----------------|-----------------|----------------|
| a            | 0.020381438517  | -0.003416414496 | -0.002332965917 | 0.020797058782 |
| b            | 0.002198745968  | 0.012902170259  | -0.034066501536 | 0.036494205131 |
| c            | -0.000630379485 | -0.024257722017 | -0.042924354571 | 0.0493085659   |

Primitive Cell3 volume: 2.842066937809649e-05 native length cubed.

## Authoritative Cell3 base cell: six boundary planes and all 27 nodes

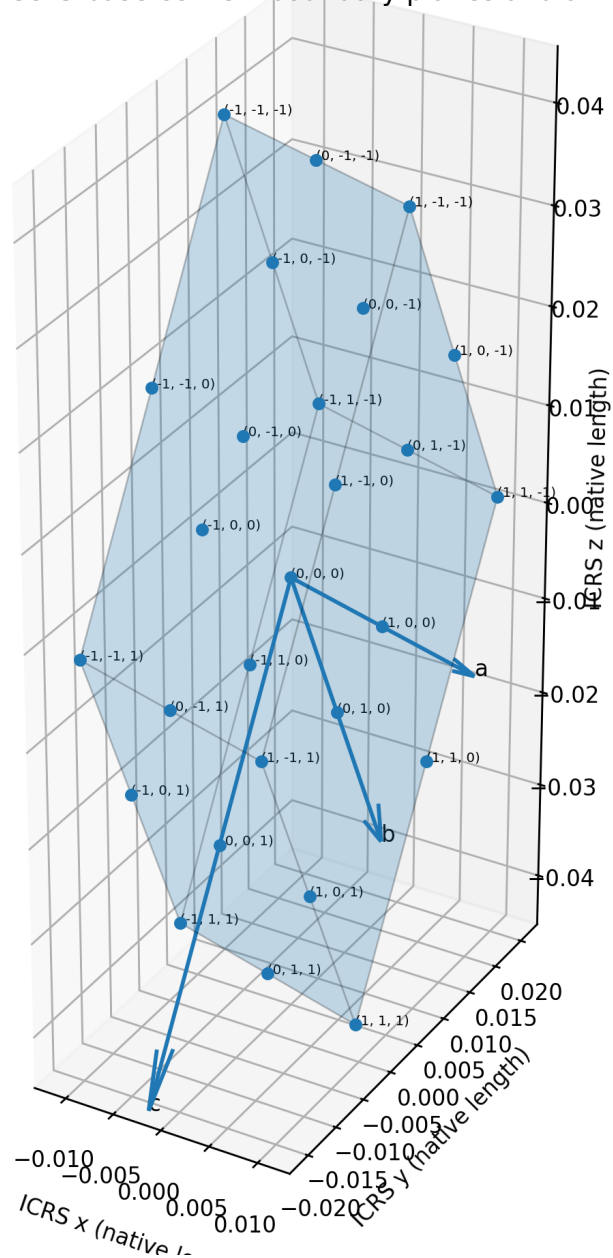


Figure 1. Authoritative Cell3 base cell in ICRS coordinates. The six translucent quadrilateral faces are the six boundary planes. Every one of the 27 centered half-step nodes is plotted and labeled by its integer half-step index (2i,2j,2k). The basis vectors a, b, and c are drawn from the origin.

Cell3 central face planes and source face normals

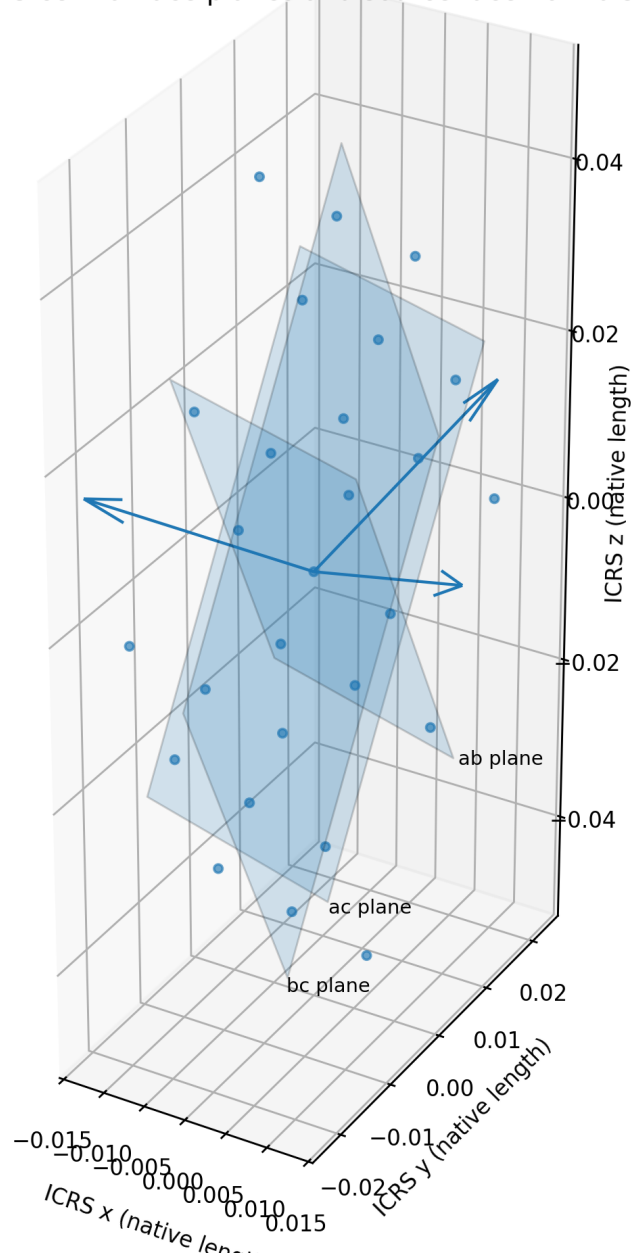


Figure 2. The three central face-plane families ab, ac, and bc. The source face normals are drawn from the origin. The three plane families and their translated positive and negative boundary members define the complete six-face Cell3 enclosure.

| Face family | Area           | Included angle | Unit normal x | Unit normal y | Unit normal z | Boundary  d    |
|-------------|----------------|----------------|---------------|---------------|---------------|----------------|
| A_ab        | 0.00075472179  | 83.933483012   | 0.1940922008  | 0.9131771688  | 0.3583792348  | 0.0188285735   |
| A_ac        | 0.001011255758 | 80.448130018   | 0.0890525941  | 0.8665767693  | -0.4910339483 | 0.014052166903 |
| A_bc        | 0.001385784432 | 50.363231814   | -0.9959651903 | 0.0836021625  | -0.0326192906 | 0.010254361619 |

Each face plane has equation  $\mathbf{n} \cdot \mathbf{x} = d$ . The opposite face uses the same normal and the negative offset. The complete vertex and node tables are included in Appendices A and B.

## 4. Quadratic phase-transition cost and gravity metric

The controlling directional operator is the symmetric positive-definite matrix  $\mathbf{Q}$ . For any direction or velocity vector  $\mathbf{x}$ , the quadratic reading is  $\mathbf{x}^T \mathbf{Q} \mathbf{x}$ . For a unit direction  $\mathbf{n}$ , the phase-transition cost is  $C(\mathbf{n}) = \mathbf{n}^T \mathbf{Q} \mathbf{n}$ . For a velocity  $\mathbf{v}$ , the  $\mathbf{Q}$ -metric kinetic energy used here is one-half  $m \kappa_F \mathbf{v}^T \mathbf{Q} \mathbf{v}$ .

$$Q = \begin{bmatrix} 2.9764106378, & 1.2431507795, & -2.0803600354 \\ 1.2431507795, & 2.5922286635, & -2.3178839937 \\ -2.0803600354, & -2.3178839937, & 6.7735709133 \end{bmatrix}$$

$$C(n) = n^T Q n$$

$$E_Q(v) = (1/2) m \kappa_F v^T Q v$$

| Mode   | Eigenvalue  | ICRS x       | ICRS y       | ICRS z       | RA (deg)   | Dec (deg)  | Ellipsoid semiaxis |
|--------|-------------|--------------|--------------|--------------|------------|------------|--------------------|
| soft   | 1.429328753 | -0.408953067 | 0.884525009  | 0.224439071  | 114.813084 | 12.969896  | 0.836438346        |
| saddle | 2.112545026 | -0.828553896 | -0.256832305 | -0.497529505 | 197.222212 | -29.836688 | 0.688013584        |
| hard   | 8.800336435 | -0.382434086 | -0.389426084 | 0.837909002  | 225.519007 | 56.91997   | 0.337093488        |

Unit-level quadratic phase-transition cost ellipsoid

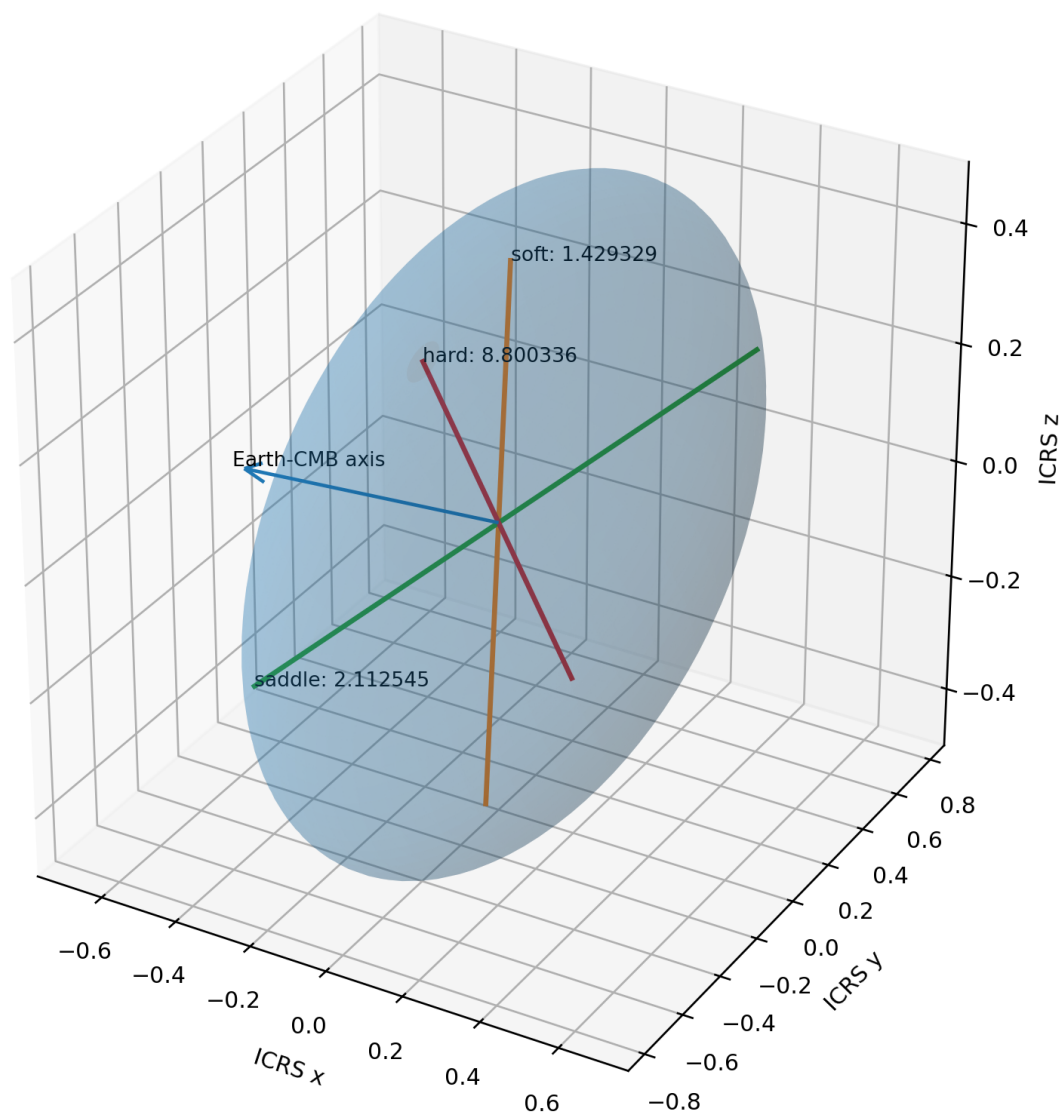


Figure 3. Unit-level ellipsoid  $x^T Q x=1$ . The soft, saddle, and hard principal axes are the eigenvectors of  $Q$ . The hard direction has the largest cost eigenvalue and the shortest ellipsoid semiaxis; the soft direction has the smallest cost eigenvalue and the longest semiaxis.

## 5. Exact 2:1 compound-rotation geometry

Let  $\phi = \omega_B t$  be the outer mechanical phase. The transverse unit vector  $e(\phi)$  rotates about the Earth-CMB axis. The inner bar phase is  $2\phi$ . The positive endpoint direction  $u(\phi)$  is the unit vector from the center to mass plus; mass minus is always at  $-u(\phi)$ .

$$\begin{aligned} e(\phi) &= \cos(\phi) e_0 + \sin(\phi) A_0 \\ f(\phi) &= de/d\phi = -\sin(\phi) e_0 + \cos(\phi) A_0 \\ u(\phi) &= \cos(2\phi) e(\phi) + \sin(2\phi) c_{\text{hat}} \\ r_{\text{plus}} &= +R u(\phi) \\ r_{\text{minus}} &= -R u(\phi) \\ u_{\phi} &= du/d\phi \\ &= -2 \sin(2\phi) e(\phi) + \cos(2\phi) f(\phi) + 2 \cos(2\phi) c_{\text{hat}} \end{aligned}$$

The fixed-frame harmonic expansion is useful for integration. It separates the trajectory into first-, second-, and third-harmonic components in the orthonormal basis  $(e_0, A_0, c_{\text{hat}})$ .

$$\begin{aligned} u(\phi) &= (1/2)[\cos(3\phi) + \cos(\phi)] e_0 \\ &\quad + (1/2)[\sin(3\phi) - \sin(\phi)] A_0 \\ &\quad + \sin(2\phi) c_{\text{hat}} \end{aligned}$$

Each labeled mass closes after one complete axis-B rotation. In that interval axis A completes two rotations, each mass reaches the CMB apex twice and the antapex twice, and the mass pair remains exactly antipodal.

## 6. Earth translation and unequal endpoint kinetic energies

The Earth-CMB velocity is common to both endpoints. Their rotational velocities are equal and opposite. The field-frame velocities are therefore not negatives of one another; they are displaced by the common translation  $V_E$ .

$$\begin{aligned} v_{\text{plus}} &= V_E + R \omega_B u_{\phi} \\ v_{\text{minus}} &= V_E - R \omega_B u_{\phi} \end{aligned}$$

The ordinary Euclidean kinetic-energy difference is controlled by the Euclidean velocity cross term. The K-Field kinetic-energy difference is controlled by the metric cross term  $u_{\phi}^T Q V_E$ . The latter is the exact advancing-retreating quantity used in the force derivation.

$$\begin{aligned} K_{\text{plus}} - K_{\text{minus}} &= 2 m R \omega_B (u_{\phi} \cdot V_E) \\ E_{Q,\text{plus}} - E_{Q,\text{minus}} \\ &= 2 m \kappa_F R \omega_B (u_{\phi}^T Q V_E) \end{aligned}$$



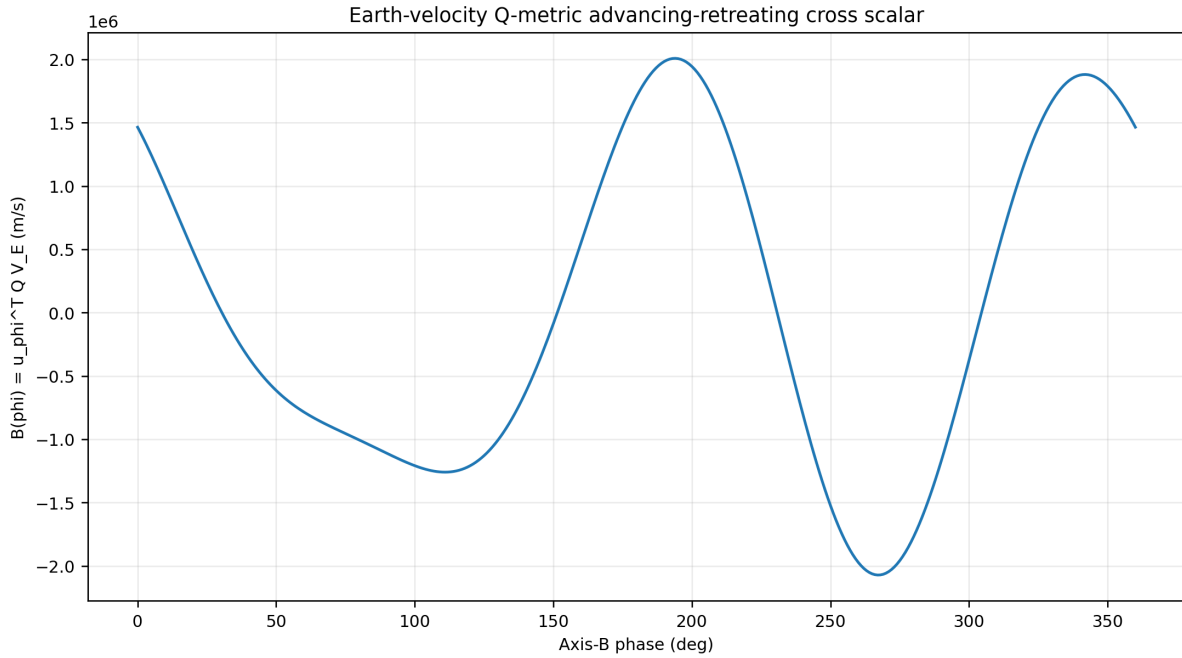


Figure 4. The exact Q-metric translation-rotation cross scalar  $B(\phi)=u_{\phi}^T Q V_E$  over one outer rotation. This term changes the individual endpoint energies and survives in the system-force sum.

## 7. Individual force vectors and system net force

The force channel is the negative radial derivative of the Q-metric kinetic energy. Radius  $R$  is the generalized radial coordinate of each endpoint measured outward from the center. The derivative is taken while holding phase and angular speed fixed.

$$\begin{aligned} E_{Q,plus} &= (1/2)m \kappa_F v_{plus}^T Q v_{plus} \\ E_{Q,minus} &= (1/2)m \kappa_F v_{minus}^T Q v_{minus} \end{aligned}$$

$$\begin{aligned} B(\phi) &= u_{\phi}^T Q V_E \\ C(\phi) &= u_{\phi}^T Q u_{\phi} \end{aligned}$$

For mass plus, the outward radial direction is  $+u$ . For mass minus, the outward radial direction is  $-u$ . Preserving those opposite local directions is essential to the vector signs.

$$F_{plus} = -\kappa_F m [\omega_B B + R \omega_B^2 C] u$$

$$F_{minus} = \kappa_F m [R \omega_B^2 C - \omega_B B] u$$

Adding the two vectors cancels the equal pure-rotation terms proportional to  $R \omega_B^2 C$ . The Earth-translation cross terms have the same global vector direction and reinforce.

### Exact system force:

$$F_{net}(\phi) = -2 \kappa_F m \omega_B [u_{\phi}^T Q V_E] u(\phi)$$

$$\begin{aligned} F_{net} &= F_{plus} + F_{minus} \\ &= -2 \kappa_F m \omega_B B(\phi) u(\phi) \end{aligned}$$

$$F_{hub} = -F_{net}$$

The hub reaction is equal and opposite to the force on the two-mass subsystem. The instantaneous system force is linear in mass,  $\kappa_F$ , and  $\omega_B$ . The individual forces additionally contain equal and opposite terms proportional to  $R \omega_B^2$ .

## 8. One-degree force time series

The complete one-degree calculation contains 361 rows from 0 through 360 degrees, including the repeated closure endpoint. Each row records the endpoint direction, its phase derivative, the Q-metric cross and rotation scalars, the two kinetic energies, the individual force vectors, the system net force, and the hub reaction.

| Quantity              | Value at $\omega_B=1$ rad/s and $\kappa_F=1$ |
|-----------------------|----------------------------------------------|
| Maximum $ F_{net} $   | 4.144924e+06 N                               |
| Phase of maximum      | 267 deg                                      |
| RMS $ F_{net} $       | 2.493223e+06 N                               |
| Mean net-force vector | (18914.376424, 1.824735e+05, 1.611087e+05) N |
| Rows                  | 361                                          |
| Increment             | 1 deg                                        |

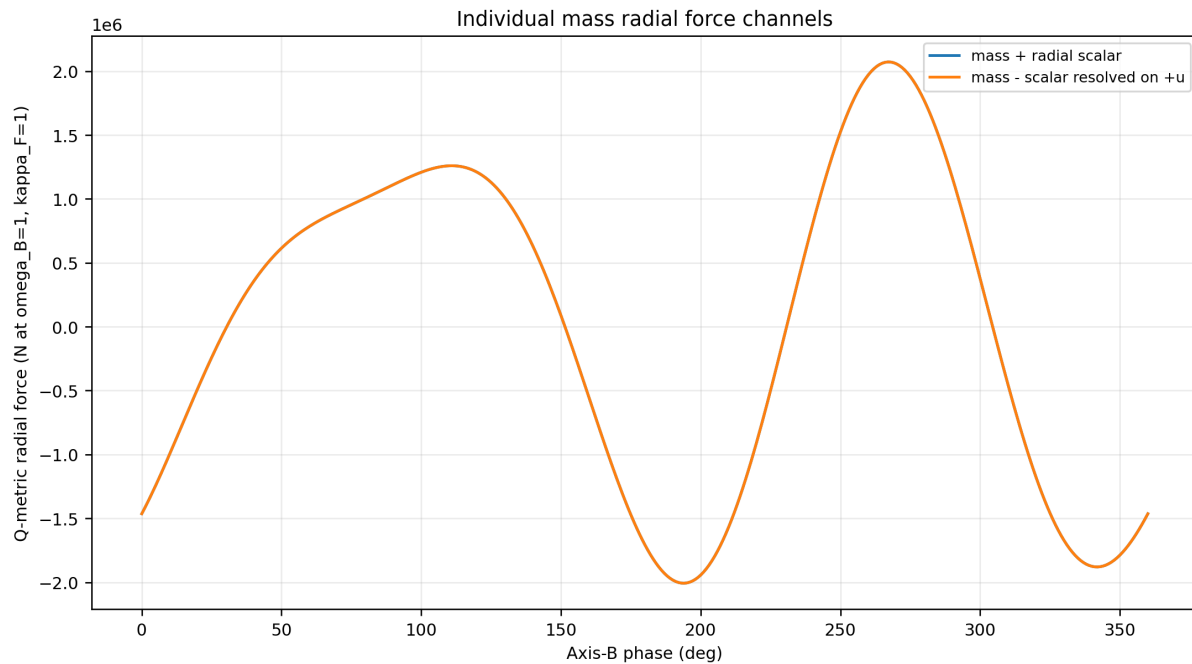


Figure 5. Individual radial force scalars. The mass-minus scalar is resolved on the global +u direction so that the two vector contributions can be compared directly. Their small pure-rotation separation is superposed on the much larger common Earth-translation cross-term force.

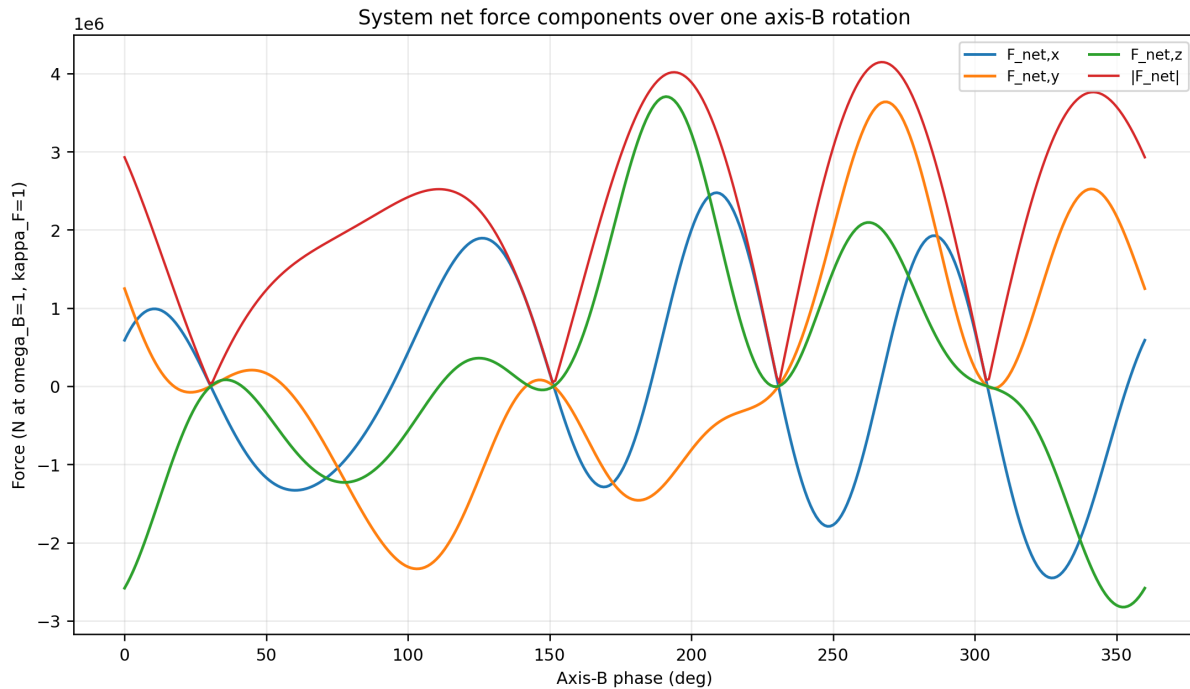


Figure 6. ICRS components and magnitude of the system net force for one complete axis-B rotation. The force is the exact vector sum of the two individual endpoint forces.

## 9. Exact closed-cycle force integration

Integrating force over time gives impulse. The outer period is  $T_B = 2\pi/\omega_B$  and  $d\phi = \omega_B dt$ . Substitution of the system force eliminates  $\omega_B$  from the cycle integral.

$$\begin{aligned} J_B &= \int_0^{T_B} F_{\text{net}}(t) dt \\ &= -2 \kappa_F m \int_0^{2\pi} [u_{\phi}^T Q V_E] u d\phi \end{aligned}$$

Define the dyadic integral  $M = \int_0^{2\pi} u u_{\phi}^T d\phi$ . The harmonic expansion of  $u$  and orthogonality of distinct sine and cosine harmonics give the following matrix in the right-handed basis ( $e_0, A_0, c_{\text{hat}}$ ).

$$\begin{aligned} M &= \int_0^{2\pi} u u_{\phi}^T d\phi \\ &= \begin{bmatrix} 0 & \pi/2 & 0 \\ -\pi/2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ &= -(\pi/2) [c_{\text{hat}} \times] \end{aligned}$$

Here  $[c_{\text{hat}} \times]$  is the skew-symmetric matrix that maps a vector  $p$  to  $c_{\text{hat}} \times p$ . With  $p = Q V_E$ , the integral closes exactly.

$$\begin{aligned} \text{Closed-cycle impulse theorem:} \\ J_B &= \pi \kappa_F m c_{\text{hat}} \times (Q V_E) \end{aligned}$$

The exact result is independent of the transverse phase  $e_0$ , the rotation speed  $\omega_B$ , and the radius  $R$ . Those quantities change the instantaneous path and force waveform, but not the closed-cycle impulse for the defined 2:1 geometry and radial force law.

| Quantity                      | x               | y               | z               | Magnitude       |
|-------------------------------|-----------------|-----------------|-----------------|-----------------|
| Analytic impulse (N s)        | 1.188425320e+05 | 1.146514978e+06 | 1.012275515e+06 | 1.534054060e+06 |
| 1-deg numerical impulse (N s) | 1.188425320e+05 | 1.146514978e+06 | 1.012275515e+06 | 1.534054060e+06 |
| Numerical minus analytic      | -2.474e-10      | 0.000e+00       | 1.048e-09       | 1.077e-09       |

| Quantity                 | x               | y               | z               | Magnitude       |
|--------------------------|-----------------|-----------------|-----------------|-----------------|
| Unit-rate mean force (N) | 18914.376424141 | 1.824735260e+05 | 1.611086520e+05 | 2.441522866e+05 |

The impulse unit direction is (0.077469585, 0.747375863, 0.659869519), corresponding to RA=84.0821143 deg and Dec=41.2899223 deg. Its dot product with the Earth-CMB unit axis is 1.294e-11 N s, establishing that the impulse is transverse to axis B.

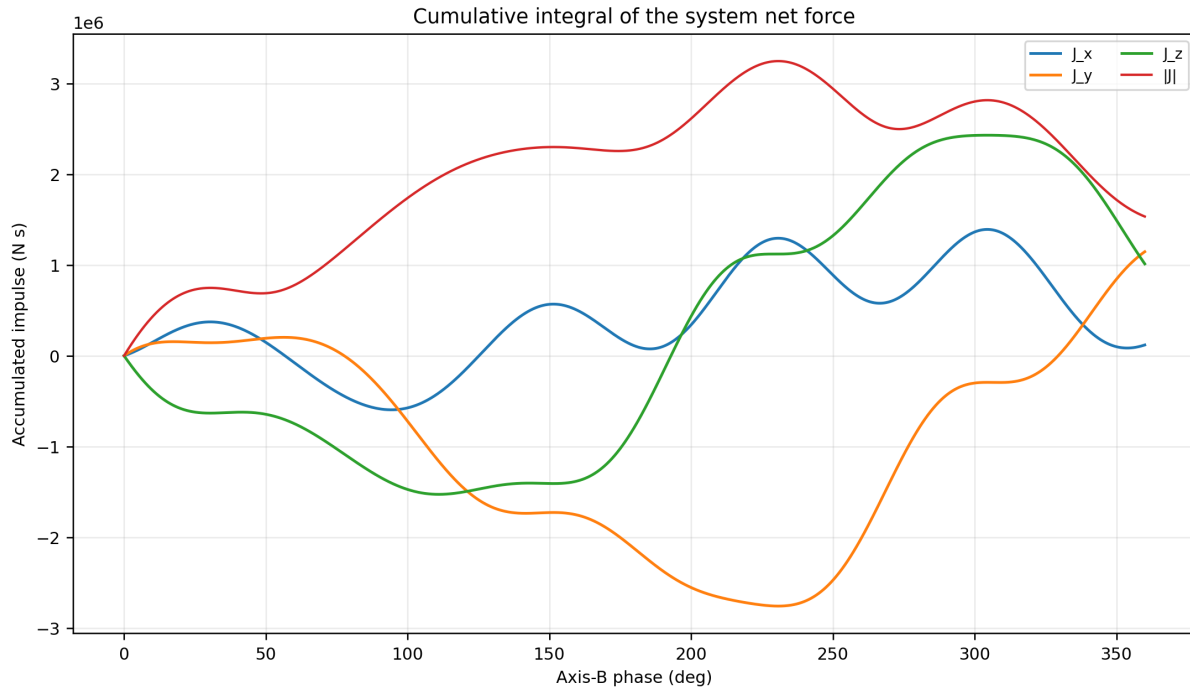


Figure 7. Cumulative impulse components and magnitude during one complete axis-B rotation. The endpoint at 360 degrees equals the exact analytic impulse vector.

## 10. Scaling laws and physical reading

The impulse theorem separates geometric direction from operating scale. The vector  $\hat{c} \text{ cross } (Q \mathbf{V}_E)$  is fixed by the Earth-CMB state and the Q operator. Mass and  $\kappa_F$  scale its magnitude. Rotation speed controls how rapidly cycles occur, but it does not change the impulse delivered by one complete cycle.

| Quantity                       | Exact scaling                                                  |
|--------------------------------|----------------------------------------------------------------|
| Instantaneous system force     | proportional to $\kappa_F m \omega_B$                          |
| Individual pure-rotation terms | proportional to $\kappa_F m R \omega_B^2$                      |
| Impulse per outer cycle        | proportional to $\kappa_F m$ ; independent of R and $\omega_B$ |
| Cycle-average force            | proportional to $\kappa_F m \omega_B$                          |
| Hub reaction                   | equal to $-F_{\text{net}}$ at every phase                      |

$$\begin{aligned}
 \langle F_{\text{net}} \rangle &= J_B / T_B \\
 &= (\omega_B / 2\pi) J_B \\
 &= (\kappa_F m \omega_B / 2) \\
 &\quad [\hat{c} \text{ cross } (Q \mathbf{V}_E)]
 \end{aligned}$$

The Q operator rotates the effective advancing-retreating direction away from the Euclidean velocity direction. The controlling projection is  $\mathbf{u}_\phi \cdot (Q \mathbf{V}_E)$ , not merely  $\mathbf{u}_\phi \cdot \mathbf{V}_E$ . The observed force direction therefore follows the anisotropic phase metric as well as the Earth translation.

For  $\kappa_F=1$  the force and impulse values are expressed in the native Q-metric mechanical normalization. A physical device with constitutive gain  $\kappa_F$  multiplies every force, impulse, and Q-metric energy difference by that gain without changing phase, direction, or harmonic structure.

## 11. Reproduction procedure

1. Load the authoritative Q matrix, Earth-CMB unit vector, and optimized 2:1 transverse reference vectors in ICRS coordinates.
2. Set  $m=1$  kg,  $R=0.5$  m,  $\omega_A=2 \omega_B$ , and the desired  $\kappa_F$ .
3. For each outer phase  $\phi$ , construct  $e(\phi)$ ,  $f(\phi)$ ,  $u(\phi)$ , and  $u_\phi$ .
4. Construct  $v_{\text{plus}}=V_E+R \omega_B u_\phi$  and  $v_{\text{minus}}=V_E-R \omega_B u_\phi$ .
5. Evaluate  $B(\phi)=u_\phi^T Q V_E$  and  $C(\phi)=u_\phi^T Q u_\phi$ .
6. Evaluate  $F_{\text{plus}}$  and  $F_{\text{minus}}$  with the exact radial derivatives and opposite local radial directions.
7. Add the individual vectors to obtain  $F_{\text{net}}=-2 \kappa_F m \omega_B B(\phi)u(\phi)$ .
8. Sample from 0 through 360 degrees in one-degree increments for the complete time series.
9. Integrate  $F_{\text{net}} dt$  with  $d\phi=\omega_B dt$  and compare with  $\pi \kappa_F m \hat{c} \text{ cross } (Q V_E)$ .
10. Generate Cell3 basis, vertices, face planes, and all 27 nodes directly from the source basis matrix.

## 12. Canonical result statement

**For the exact 2:1 dual-axis apparatus with axis B aligned to the Earth-CMB velocity, axis A perpendicular to axis B, two 1 kg endpoint masses at radius 0.5 m, and Q-metric radial force derived from one-half  $m \kappa_F v^T Q v$ , the system net force is:**

$$F_{\text{net}}(\phi) = -2 \kappa_F m \omega_B [u_\phi^T Q V_E] u(\phi).$$

**The complete impulse over one axis-B rotation is:**

$$J_B = \pi \kappa_F m \hat{c} \text{ cross } (Q V_E).$$

**At  $\kappa_F=1$  and  $m=1$  kg,  $J_B=(118842.532043, 1146514.977540, 1012275.514802)$  N s, with magnitude  $1.534054060 \times 10^6$  N s. The impulse is exactly transverse to the Earth-CMB axis. The one-degree numerical integration reproduces the analytic vector to approximately  $10^{-9}$  N s.**

## Appendix A. Complete Cell3 boundary-vertex table

| # | i    | j    | k    | ICRS x          | ICRS y          | ICRS z          |
|---|------|------|------|-----------------|-----------------|-----------------|
| 0 | -0.5 | -0.5 | -0.5 | -0.0109749025   | 0.007385983127  | 0.039661911012  |
| 1 | -0.5 | -0.5 | 0.5  | -0.011605281985 | -0.01687173889  | -0.003262443559 |
| 2 | -0.5 | 0.5  | -0.5 | -0.008776156532 | 0.020288153386  | 0.005595409476  |
| 3 | -0.5 | 0.5  | 0.5  | -0.009406536017 | -0.003969568631 | -0.037328945095 |
| 4 | 0.5  | -0.5 | -0.5 | 0.009406536017  | 0.003969568631  | 0.037328945095  |
| 5 | 0.5  | -0.5 | 0.5  | 0.008776156532  | -0.020288153386 | -0.005595409476 |
| 6 | 0.5  | 0.5  | -0.5 | 0.011605281985  | 0.01687173889   | 0.003262443559  |
| 7 | 0.5  | 0.5  | 0.5  | 0.0109749025    | -0.007385983127 | -0.039661911012 |

## Appendix B. Complete 27-node centered half-step registry

| #  | 2i | 2j | 2k | lambda | ICRS x          | ICRS y          | ICRS z          |
|----|----|----|----|--------|-----------------|-----------------|-----------------|
| 0  | -1 | -1 | -1 | -3     | -0.0109749025   | 0.007385983127  | 0.039661911012  |
| 1  | -1 | -1 | 0  | -2     | -0.011290092243 | -0.004742877881 | 0.018199733726  |
| 2  | -1 | -1 | 1  | -1     | -0.011605281985 | -0.01687173889  | -0.003262443559 |
| 3  | -1 | 0  | -1 | -2     | -0.009875529516 | 0.013837068257  | 0.022628660244  |
| 4  | -1 | 0  | 0  | -1     | -0.010190719258 | 0.001708207248  | 0.001166482958  |
| 5  | -1 | 0  | 1  | 0      | -0.010505909001 | -0.01042065376  | -0.020295694327 |
| 6  | -1 | 1  | -1 | -1     | -0.008776156532 | 0.020288153386  | 0.005595409476  |
| 7  | -1 | 1  | 0  | 0      | -0.009091346274 | 0.008159292378  | -0.01586676781  |
| 8  | -1 | 1  | 1  | 1      | -0.009406536017 | -0.003969568631 | -0.037328945095 |
| 9  | 0  | -1 | -1 | -2     | -0.000784183241 | 0.005677775879  | 0.038495428054  |
| 10 | 0  | -1 | 0  | -1     | -0.001099372984 | -0.006451085129 | 0.017033250768  |
| 11 | 0  | -1 | 1  | 0      | -0.001414562727 | -0.018579946138 | -0.004428926518 |
| 12 | 0  | 0  | -1 | -1     | 0.000315189743  | 0.012128861009  | 0.021462177286  |
| 13 | 0  | 0  | 0  | 0      | 0               | 0               | 0               |
| 14 | 0  | 0  | 1  | 1      | -0.000315189743 | -0.012128861009 | -0.021462177286 |
| 15 | 0  | 1  | -1 | 0      | 0.001414562727  | 0.018579946138  | 0.004428926518  |
| 16 | 0  | 1  | 0  | 1      | 0.001099372984  | 0.006451085129  | -0.017033250768 |
| 17 | 0  | 1  | 1  | 2      | 0.000784183241  | -0.005677775879 | -0.038495428054 |
| 18 | 1  | -1 | -1 | -1     | 0.009406536017  | 0.003969568631  | 0.037328945095  |
| 19 | 1  | -1 | 0  | 0      | 0.009091346274  | -0.008159292378 | 0.01586676781   |
| 20 | 1  | -1 | 1  | 1      | 0.008776156532  | -0.020288153386 | -0.005595409476 |
| 21 | 1  | 0  | -1 | 0      | 0.010505909001  | 0.01042065376   | 0.020295694327  |
| 22 | 1  | 0  | 0  | 1      | 0.010190719258  | -0.001708207248 | -0.001166482958 |
| 23 | 1  | 0  | 1  | 2      | 0.009875529516  | -0.013837068257 | -0.022628660244 |
| 24 | 1  | 1  | -1 | 1      | 0.011605281985  | 0.01687173889   | 0.003262443559  |
| 25 | 1  | 1  | 0  | 2      | 0.011290092243  | 0.004742877881  | -0.018199733726 |
| 26 | 1  | 1  | 1  | 3      | 0.0109749025    | -0.007385983127 | -0.039661911012 |

## Appendix C. Complete one-degree individual and system force vectors

All values use omega\_B=1 rad/s, omega\_A=2 rad/s, kappa\_F=1, m=1 kg, and R=0.5 m. The repeated 360-degree row is the closure endpoint. Scientific notation is used to preserve width and precision.

| phase | F+ x         | F+ y         | F+ z         | F- x         | F- y         | F- z         | Fnet x       | Fnet y       | Fnet z       | Fnet        |
|-------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|
| 0     | 2.94040e+05  | 6.23948e+05  | -1.29118e+06 | 2.94038e+05  | 6.23944e+05  | -1.29117e+06 | 5.88079e+05  | 1.24789e+06  | -2.58235e+06 | 2.92773e+06 |
| 1     | 3.30303e+05  | 5.73462e+05  | -1.25848e+06 | 3.30301e+05  | 5.73458e+05  | -1.25848e+06 | 6.60603e+05  | 1.14692e+06  | -2.51696e+06 | 2.84375e+06 |
| 2     | 3.63104e+05  | 5.23590e+05  | -1.22245e+06 | 3.63102e+05  | 5.23587e+05  | -1.22244e+06 | 7.26206e+05  | 1.04718e+06  | -2.44489e+06 | 2.75707e+06 |
| 3     | 3.92352e+05  | 4.74636e+05  | -1.18331e+06 | 3.92350e+05  | 4.74634e+05  | -1.18330e+06 | 7.84701e+05  | 9.49270e+05  | -2.36661e+06 | 2.66791e+06 |
| 4     | 4.17976e+05  | 4.26886e+05  | -1.14132e+06 | 4.17973e+05  | 4.26884e+05  | -1.14131e+06 | 8.35949e+05  | 8.53770e+05  | -2.28263e+06 | 2.57646e+06 |
| 5     | 4.39927e+05  | 3.80607e+05  | -1.09675e+06 | 4.39924e+05  | 3.80605e+05  | -1.09675e+06 | 8.79852e+05  | 7.61212e+05  | -2.19350e+06 | 2.48295e+06 |
| 6     | 4.58180e+05  | 3.36045e+05  | -1.04990e+06 | 4.58177e+05  | 3.36043e+05  | -1.04989e+06 | 9.16357e+05  | 6.72088e+05  | -2.09979e+06 | 2.38758e+06 |
| 7     | 4.72728e+05  | 2.93425e+05  | -1.00105e+06 | 4.72725e+05  | 2.93423e+05  | -1.00104e+06 | 9.45453e+05  | 5.86848e+05  | -2.00209e+06 | 2.29055e+06 |
| 8     | 4.83588e+05  | 2.52947e+05  | -9.50516e+05 | 4.83584e+05  | 2.52945e+05  | -9.50508e+05 | 9.67172e+05  | 5.05892e+05  | -1.90102e+06 | 2.19209e+06 |
| 9     | 4.90794e+05  | 2.14786e+05  | -8.98618e+05 | 4.90789e+05  | 2.14784e+05  | -8.98610e+05 | 9.81583e+05  | 4.29570e+05  | -1.79723e+06 | 2.09238e+06 |
| 10    | 4.94399e+05  | 1.79093e+05  | -8.45676e+05 | 4.94395e+05  | 1.79091e+05  | -8.45668e+05 | 9.88794e+05  | 3.58184e+05  | -1.69134e+06 | 1.99165e+06 |
| 11    | 4.94478e+05  | 1.45991e+05  | -7.92012e+05 | 4.94472e+05  | 1.45990e+05  | -7.92004e+05 | 9.88950e+05  | 2.91981e+05  | -1.58402e+06 | 1.89007e+06 |
| 12    | 4.91118e+05  | 1.15579e+05  | -7.37952e+05 | 4.91112e+05  | 1.15578e+05  | -7.37943e+05 | 9.82230e+05  | 2.31157e+05  | -1.47589e+06 | 1.78787e+06 |
| 13    | 4.84425e+05  | 8.79274e+04  | -6.83814e+05 | 4.84418e+05  | 8.79261e+04  | -6.83805e+05 | 9.68844e+05  | 1.75854e+05  | -1.36762e+06 | 1.68522e+06 |
| 14    | 4.74520e+05  | 6.30812e+04  | -6.29915e+05 | 4.74512e+05  | 6.30802e+04  | -6.29905e+05 | 9.49032e+05  | 1.26161e+05  | -1.25982e+06 | 1.58232e+06 |
| 15    | 4.61536e+05  | 4.10597e+04  | -5.76563e+05 | 4.61528e+05  | 4.10590e+04  | -5.76553e+05 | 9.23064e+05  | 8.21187e+04  | -1.15312e+06 | 1.47935e+06 |
| 16    | 4.45620e+05  | 2.18570e+04  | -5.24056e+05 | 4.45611e+05  | 2.18566e+04  | -5.24045e+05 | 8.91230e+05  | 4.37136e+04  | -1.04810e+06 | 1.37649e+06 |
| 17    | 4.26929e+05  | 5.44283e+03  | -4.72680e+05 | 4.26919e+05  | 5.44270e+03  | -4.72669e+05 | 8.53847e+05  | 1.08855e+04  | -9.45349e+05 | 1.27391e+06 |
| 18    | 4.05630e+05  | -8.23624e+03 | -4.22708e+05 | 4.05619e+05  | -8.23601e+03 | -4.22697e+05 | 8.11249e+05  | -1.64722e+04 | -8.45405e+05 | 1.17180e+06 |
| 19    | 3.81900e+05  | -1.92556e+04 | -3.74397e+05 | 3.81888e+05  | -1.92550e+04 | -3.74386e+05 | 7.63788e+05  | -3.85105e+04 | -7.48783e+05 | 1.07030e+06 |
| 20    | 3.55921e+05  | -2.77111e+04 | -3.27987e+05 | 3.55908e+05  | -2.77101e+04 | -3.27975e+05 | 7.11830e+05  | -5.54212e+04 | -6.55963e+05 | 9.69567e+05 |
| 21    | 3.27883e+05  | -3.37179e+04 | -2.83699e+05 | 3.27869e+05  | -3.37164e+04 | -2.83687e+05 | 6.55752e+05  | -6.74343e+04 | -5.67387e+05 | 8.69762e+05 |
| 22    | 2.97978e+05  | -3.74080e+04 | -2.41735e+05 | 2.97963e+05  | -3.74062e+04 | -2.41723e+05 | 5.95941e+05  | -7.48142e+04 | -4.83458e+05 | 7.71022e+05 |
| 23    | 2.66403e+05  | -3.89295e+04 | -2.02275e+05 | 2.66387e+05  | -3.89272e+04 | -2.02263e+05 | 5.32790e+05  | -7.78567e+04 | -4.04537e+05 | 6.73481e+05 |
| 24    | 2.33357e+05  | -3.84439e+04 | -1.65477e+05 | 2.33340e+05  | -3.84411e+04 | -1.65465e+05 | 4.66696e+05  | -7.68850e+04 | -3.30942e+05 | 5.77269e+05 |
| 25    | 1.99038e+05  | -3.61244e+04 | -1.31479e+05 | 1.99020e+05  | -3.61212e+04 | -1.31467e+05 | 3.98059e+05  | -7.22456e+04 | -2.62946e+05 | 4.82505e+05 |
| 26    | 1.63647e+05  | -3.21543e+04 | -1.00394e+05 | 1.63628e+05  | -3.21506e+04 | -1.00383e+05 | 3.27274e+05  | -6.43050e+04 | -2.00777e+05 | 3.89300e+05 |
| 27    | 1.27379e+05  | -2.67246e+04 | -7.23133e+04 | 1.27359e+05  | -2.67205e+04 | -7.23021e+04 | 2.54738e+05  | -5.34451e+04 | -1.44615e+05 | 2.97760e+05 |
| 28    | 9.04290e+04  | -2.00322e+04 | -4.73045e+04 | 9.04085e+04  | -2.00277e+04 | -4.72938e+04 | 1.80837e+05  | -4.00599e+04 | -9.45983e+04 | 2.07980e+05 |
| 29    | 5.29885e+04  | -1.22780e+04 | -2.54132e+04 | 5.29673e+04  | -1.22731e+04 | -2.54030e+04 | 1.05956e+05  | -2.45511e+04 | -5.08162e+04 | 1.20049e+05 |
| 30    | 1.52433e+04  | -3.66476e+03 | -6.66229e+03 | 1.52214e+04  | -3.65951e+03 | -6.65274e+03 | 3.04647e+04  | -7.32427e+03 | -1.33150e+04 | 3.40446e+04 |
| 31    | -2.26260e+04 | 5.60437e+03  | 8.94688e+03  | -2.26484e+04 | 5.60992e+03  | 8.95574e+03  | -4.52745e+04 | 1.12143e+04  | 1.79026e+04  | 4.99604e+04 |
| 32    | -6.04452e+04 | 1.53280e+04  | 2.14340e+04  | -6.04680e+04 | 1.53338e+04  | 2.14422e+04  | -1.20913e+05 | 3.06617e+04  | 4.28762e+04  | 1.31903e+05 |
| 33    | -9.80469e+04 | 2.53078e+04  | 3.08390e+04  | -9.80702e+04 | 2.53138e+04  | 3.08463e+04  | -1.96117e+05 | 5.06216e+04  | 6.16852e+04  | 2.11730e+05 |
| 34    | -1.35272e+05 | 3.53505e+04  | 3.72205e+04  | -1.35295e+05 | 3.53567e+04  | 3.72270e+04  | -2.70567e+05 | 7.07072e+04  | 7.44475e+04  | 2.89393e+05 |
| 35    | -1.71968e+05 | 4.52691e+04  | 4.06554e+04  | -1.71992e+05 | 4.52753e+04  | 4.06610e+04  | -3.43960e+05 | 9.05445e+04  | 8.13163e+04  | 3.64855e+05 |
| 36    | -2.07994e+05 | 5.48841e+04  | 4.12369e+04  | -2.08018e+05 | 5.48904e+04  | 4.12416e+04  | -4.16012e+05 | 1.09775e+05  | 8.24785e+04  | 4.38086e+05 |
| 37    | -2.43216e+05 | 6.40250e+04  | 3.90736e+04  | -2.43240e+05 | 6.40312e+04  | 3.90774e+04  | -4.86456e+05 | 1.28056e+05  | 7.81510e+04  | 5.09064e+05 |
| 38    | -2.77511e+05 | 7.25310e+04  | 3.42880e+04  | -2.77534e+05 | 7.25371e+04  | 3.42909e+04  | -5.55045e+05 | 1.45068e+05  | 6.85788e+04  | 5.77774e+05 |
| 39    | -3.10763e+05 | 8.02522e+04  | 2.70146e+04  | -3.10786e+05 | 8.02582e+04  | 2.70166e+04  | -6.21549e+05 | 1.60510e+05  | 5.40311e+04  | 6.44210e+05 |

| phase | F+ x         | F+ y         | F+ z         | F- x         | F- y         | F- z         | Fnet x       | Fnet y       | Fnet z       | Fnet        |
|-------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|
| 40    | -3.42869e+05 | 8.70502e+04  | 1.73988e+04  | -3.42892e+05 | 8.70560e+04  | 1.74000e+04  | -6.85760e+05 | 1.74106e+05  | 3.47988e+04  | 7.08372e+05 |
| 41    | -3.73734e+05 | 9.27987e+04  | 5.59528e+03  | -3.73757e+05 | 9.28043e+04  | 5.59561e+03  | -7.47491e+05 | 1.85603e+05  | 1.11909e+04  | 7.70270e+05 |
| 42    | -4.03275e+05 | 9.73841e+04  | -8.23384e+03 | -4.03297e+05 | 9.73893e+04  | -8.23429e+03 | -8.06571e+05 | 1.94773e+05  | -1.64681e+04 | 8.29919e+05 |
| 43    | -4.31416e+05 | 1.00705e+05  | -2.39203e+04 | -4.31437e+05 | 1.00710e+05  | -2.39215e+04 | -8.62853e+05 | 2.01416e+05  | -4.78419e+04 | 8.87340e+05 |
| 44    | -4.58093e+05 | 1.02675e+05  | -4.12915e+04 | -4.58114e+05 | 1.02680e+05  | -4.12934e+04 | -9.16207e+05 | 2.05355e+05  | -8.25849e+04 | 9.42563e+05 |
| 45    | -4.83251e+05 | 1.03218e+05  | -6.01717e+04 | -4.83271e+05 | 1.03222e+05  | -6.01741e+04 | -9.66521e+05 | 2.06440e+05  | -1.20346e+05 | 9.95623e+05 |
| 46    | -5.06843e+05 | 1.02273e+05  | -8.03836e+04 | -5.06862e+05 | 1.02277e+05  | -8.03866e+04 | -1.01371e+06 | 2.04550e+05  | -1.60770e+05 | 1.04656e+06 |
| 47    | -5.28834e+05 | 9.97904e+04  | -1.01750e+05 | -5.28852e+05 | 9.97938e+04  | -1.01753e+05 | -1.05769e+06 | 1.99584e+05  | -2.03503e+05 | 1.09542e+06 |
| 48    | -5.49193e+05 | 9.57338e+04  | -1.24094e+05 | -5.49211e+05 | 9.57368e+04  | -1.24098e+05 | -1.09840e+06 | 1.91471e+05  | -2.48192e+05 | 1.14226e+06 |
| 49    | -5.67902e+05 | 9.00784e+04  | -1.47242e+05 | -5.67919e+05 | 9.00809e+04  | -1.47247e+05 | -1.13582e+06 | 1.80159e+05  | -2.94489e+05 | 1.18713e+06 |
| 50    | -5.84948e+05 | 8.28107e+04  | -1.71024e+05 | -5.84963e+05 | 8.28128e+04  | -1.71029e+05 | -1.16991e+06 | 1.65624e+05  | -3.42053e+05 | 1.23009e+06 |
| 51    | -6.00325e+05 | 7.39281e+04  | -1.95274e+05 | -6.00339e+05 | 7.39298e+04  | -1.95278e+05 | -1.20066e+06 | 1.47858e+05  | -3.90552e+05 | 1.27121e+06 |
| 52    | -6.14034e+05 | 6.34380e+04  | -2.19830e+05 | -6.14048e+05 | 6.34393e+04  | -2.19835e+05 | -1.22808e+06 | 1.26877e+05  | -4.39664e+05 | 1.31057e+06 |
| 53    | -6.26083e+05 | 5.13571e+04  | -2.44538e+05 | -6.26096e+05 | 5.13581e+04  | -2.44543e+05 | -1.25218e+06 | 1.02715e+05  | -4.89081e+05 | 1.34822e+06 |
| 54    | -6.36485e+05 | 3.77109e+04  | -2.69251e+05 | -6.36497e+05 | 3.77116e+04  | -2.69256e+05 | -1.27298e+06 | 7.54225e+04  | -5.38507e+05 | 1.38426e+06 |
| 55    | -6.45258e+05 | 2.25324e+04  | -2.93828e+05 | -6.45269e+05 | 2.25328e+04  | -2.93833e+05 | -1.29053e+06 | 4.50652e+04  | -5.87660e+05 | 1.41874e+06 |
| 56    | -6.52424e+05 | 5.86184e+03  | -3.18136e+05 | -6.52434e+05 | 5.86193e+03  | -3.18141e+05 | -1.30486e+06 | 1.17238e+04  | -6.36277e+05 | 1.45177e+06 |
| 57    | -6.58008e+05 | -1.22545e+04 | -3.42051e+05 | -6.58017e+05 | -1.22547e+04 | -3.42055e+05 | -1.31603e+06 | -2.45092e+04 | -6.84106e+05 | 1.48342e+06 |
| 58    | -6.62040e+05 | -3.17650e+04 | -3.65456e+05 | -6.62049e+05 | -3.17654e+04 | -3.65460e+05 | -1.32409e+06 | -6.35304e+04 | -7.30916e+05 | 1.51376e+06 |
| 59    | -6.64551e+05 | -5.26132e+04 | -3.88243e+05 | -6.64559e+05 | -5.26139e+04 | -3.88248e+05 | -1.32911e+06 | -1.05227e+05 | -7.76491e+05 | 1.54290e+06 |
| 60    | -6.65575e+05 | -7.47390e+04 | -4.10313e+05 | -6.65582e+05 | -7.47398e+04 | -4.10318e+05 | -1.33116e+06 | -1.49479e+05 | -8.20631e+05 | 1.57091e+06 |
| 61    | -6.65147e+05 | -9.80789e+04 | -4.31575e+05 | -6.65153e+05 | -9.80799e+04 | -4.31580e+05 | -1.33030e+06 | -1.96159e+05 | -8.63155e+05 | 1.59788e+06 |
| 62    | -6.63302e+05 | -1.22567e+05 | -4.51946e+05 | -6.63308e+05 | -1.22568e+05 | -4.51950e+05 | -1.32661e+06 | -2.45135e+05 | -9.03896e+05 | 1.62389e+06 |
| 63    | -6.60076e+05 | -1.48135e+05 | -4.71349e+05 | -6.60082e+05 | -1.48136e+05 | -4.71354e+05 | -1.32016e+06 | -2.96271e+05 | -9.42703e+05 | 1.64903e+06 |
| 64    | -6.55507e+05 | -1.74713e+05 | -4.89719e+05 | -6.55512e+05 | -1.74715e+05 | -4.89723e+05 | -1.31102e+06 | -3.49428e+05 | -9.79441e+05 | 1.67337e+06 |
| 65    | -6.49628e+05 | -2.02232e+05 | -5.06993e+05 | -6.49633e+05 | -2.02233e+05 | -5.06997e+05 | -1.29926e+06 | -4.04465e+05 | -1.01399e+06 | 1.69701e+06 |
| 66    | -6.42475e+05 | -2.30619e+05 | -5.23118e+05 | -6.42480e+05 | -2.30620e+05 | -5.23122e+05 | -1.28496e+06 | -4.61239e+05 | -1.04624e+06 | 1.72002e+06 |
| 67    | -6.34081e+05 | -2.59802e+05 | -5.38048e+05 | -6.34086e+05 | -2.59804e+05 | -5.38052e+05 | -1.26817e+06 | -5.19606e+05 | -1.07610e+06 | 1.74248e+06 |
| 68    | -6.24478e+05 | -2.89711e+05 | -5.51740e+05 | -6.24482e+05 | -2.89713e+05 | -5.51744e+05 | -1.24896e+06 | -5.79424e+05 | -1.10348e+06 | 1.76446e+06 |
| 69    | -6.13695e+05 | -3.20272e+05 | -5.64159e+05 | -6.13699e+05 | -3.20275e+05 | -5.64163e+05 | -1.22739e+06 | -6.40547e+05 | -1.12832e+06 | 1.78603e+06 |
| 70    | -6.01760e+05 | -3.51414e+05 | -5.75274e+05 | -6.01764e+05 | -3.51417e+05 | -5.75278e+05 | -1.20352e+06 | -7.02831e+05 | -1.15055e+06 | 1.80727e+06 |
| 71    | -5.88699e+05 | -3.83065e+05 | -5.85058e+05 | -5.88703e+05 | -3.83068e+05 | -5.85062e+05 | -1.17740e+06 | -7.66133e+05 | -1.17012e+06 | 1.82823e+06 |
| 72    | -5.74534e+05 | -4.15153e+05 | -5.93488e+05 | -5.74538e+05 | -4.15156e+05 | -5.93493e+05 | -1.14907e+06 | -8.30308e+05 | -1.18698e+06 | 1.84897e+06 |
| 73    | -5.59287e+05 | -4.47604e+05 | -6.00548e+05 | -5.59291e+05 | -4.47608e+05 | -6.00552e+05 | -1.11858e+06 | -8.95212e+05 | -1.20110e+06 | 1.86956e+06 |
| 74    | -5.42976e+05 | -4.80347e+05 | -6.06220e+05 | -5.42980e+05 | -4.80351e+05 | -6.06225e+05 | -1.08596e+06 | -9.60698e+05 | -1.21244e+06 | 1.89004e+06 |
| 75    | -5.25617e+05 | -5.13308e+05 | -6.10493e+05 | -5.25622e+05 | -5.13312e+05 | -6.10498e+05 | -1.05124e+06 | -1.02662e+06 | -1.22099e+06 | 1.91046e+06 |
| 76    | -5.07225e+05 | -5.46412e+05 | -6.13358e+05 | -5.07229e+05 | -5.46417e+05 | -6.13363e+05 | -1.01445e+06 | -1.09283e+06 | -1.22672e+06 | 1.93086e+06 |
| 77    | -4.87810e+05 | -5.79585e+05 | -6.14808e+05 | -4.87814e+05 | -5.79590e+05 | -6.14813e+05 | -9.75624e+05 | -1.15917e+06 | -1.22962e+06 | 1.95128e+06 |
| 78    | -4.67383e+05 | -6.12749e+05 | -6.14838e+05 | -4.67388e+05 | -6.12755e+05 | -6.14843e+05 | -9.34771e+05 | -1.22550e+06 | -1.22968e+06 | 1.97174e+06 |
| 79    | -4.45954e+05 | -6.45826e+05 | -6.13446e+05 | -4.45958e+05 | -6.45832e+05 | -6.13452e+05 | -8.91911e+05 | -1.29166e+06 | -1.22690e+06 | 1.99228e+06 |
| 80    | -4.23528e+05 | -6.78736e+05 | -6.10633e+05 | -4.23532e+05 | -6.78742e+05 | -6.10639e+05 | -8.47060e+05 | -1.35748e+06 | -1.22127e+06 | 2.01290e+06 |
| 81    | -4.00113e+05 | -7.11395e+05 | -6.06402e+05 | -4.00117e+05 | -7.11402e+05 | -6.06408e+05 | -8.00230e+05 | -1.42280e+06 | -1.21281e+06 | 2.03362e+06 |
| 82    | -3.75715e+05 | -7.43718e+05 | -6.00758e+05 | -3.75718e+05 | -7.43725e+05 | -6.00764e+05 | -7.51433e+05 | -1.48744e+06 | -1.20152e+06 | 2.05446e+06 |
| 83    | -3.50339e+05 | -7.75617e+05 | -5.93708e+05 | -3.50343e+05 | -7.75625e+05 | -5.93714e+05 | -7.00682e+05 | -1.55124e+06 | -1.18742e+06 | 2.07540e+06 |
| 84    | -3.23993e+05 | -8.07001e+05 | -5.85265e+05 | -3.23996e+05 | -8.07009e+05 | -5.85271e+05 | -6.47988e+05 | -1.61401e+06 | -1.17054e+06 | 2.09644e+06 |



| phase | F+ x         | F+ y         | F+ z         | F- x         | F- y         | F- z         | Fnet x       | Fnet y       | Fnet z       | Fnet        |
|-------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|
| 85    | -2.96682e+05 | -8.37776e+05 | -5.75440e+05 | -2.96685e+05 | -8.37785e+05 | -5.75446e+05 | -5.93368e+05 | -1.67556e+06 | -1.15089e+06 | 2.11758e+06 |
| 86    | -2.68417e+05 | -8.67845e+05 | -5.64253e+05 | -2.68420e+05 | -8.67854e+05 | -5.64259e+05 | -5.36837e+05 | -1.73570e+06 | -1.12851e+06 | 2.13878e+06 |
| 87    | -2.39208e+05 | -8.97109e+05 | -5.51724e+05 | -2.39210e+05 | -8.97118e+05 | -5.51730e+05 | -4.78418e+05 | -1.79423e+06 | -1.10345e+06 | 2.16003e+06 |
| 88    | -2.09067e+05 | -9.25463e+05 | -5.37879e+05 | -2.09069e+05 | -9.25473e+05 | -5.37884e+05 | -4.18137e+05 | -1.85094e+06 | -1.07576e+06 | 2.18130e+06 |
| 89    | -1.78012e+05 | -9.52803e+05 | -5.22747e+05 | -1.78014e+05 | -9.52814e+05 | -5.22752e+05 | -3.56025e+05 | -1.90562e+06 | -1.04550e+06 | 2.20254e+06 |
| 90    | -1.46061e+05 | -9.79022e+05 | -5.06363e+05 | -1.46062e+05 | -9.79032e+05 | -5.06369e+05 | -2.92123e+05 | -1.95805e+06 | -1.01273e+06 | 2.22372e+06 |
| 91    | -1.13238e+05 | -1.00401e+06 | -4.88768e+05 | -1.13240e+05 | -1.00402e+06 | -4.88773e+05 | -2.26478e+05 | -2.00803e+06 | -9.77541e+05 | 2.24478e+06 |
| 92    | -7.95727e+04 | -1.02765e+06 | -4.70007e+05 | -7.95736e+04 | -1.02766e+06 | -4.70012e+05 | -1.59146e+05 | -2.05531e+06 | -9.40020e+05 | 2.26567e+06 |
| 93    | -4.50974e+04 | -1.04984e+06 | -4.50133e+05 | -4.50979e+04 | -1.04985e+06 | -4.50137e+05 | -9.01952e+04 | -2.09968e+06 | -9.00270e+05 | 2.28633e+06 |
| 94    | -9.85129e+03 | -1.07045e+06 | -4.29202e+05 | -9.85139e+03 | -1.07047e+06 | -4.29207e+05 | -1.97027e+04 | -2.14092e+06 | -8.58409e+05 | 2.30668e+06 |
| 95    | 2.61205e+04  | -1.08939e+06 | -4.07281e+05 | 2.61207e+04  | -1.08940e+06 | -4.07286e+05 | 5.22412e+04  | -2.17879e+06 | -8.14567e+05 | 2.32667e+06 |
| 96    | 6.27665e+04  | -1.10653e+06 | -3.84441e+05 | 6.27671e+04  | -1.10654e+06 | -3.84445e+05 | 1.25534e+05  | -2.21308e+06 | -7.68886e+05 | 2.34620e+06 |
| 97    | 1.00029e+05  | -1.12177e+06 | -3.60760e+05 | 1.00030e+05  | -1.12178e+06 | -3.60764e+05 | 2.00058e+05  | -2.24356e+06 | -7.21524e+05 | 2.36520e+06 |
| 98    | 1.37841e+05  | -1.13500e+06 | -3.36324e+05 | 1.37843e+05  | -1.13502e+06 | -3.36327e+05 | 2.75684e+05  | -2.27002e+06 | -6.72650e+05 | 2.38358e+06 |
| 99    | 1.76132e+05  | -1.14612e+06 | -3.11224e+05 | 1.76134e+05  | -1.14613e+06 | -3.11227e+05 | 3.52266e+05  | -2.29226e+06 | -6.22452e+05 | 2.40125e+06 |
| 100   | 2.14821e+05  | -1.15503e+06 | -2.85562e+05 | 2.14823e+05  | -1.15505e+06 | -2.85564e+05 | 4.29644e+05  | -2.31008e+06 | -5.71126e+05 | 2.41811e+06 |
| 101   | 2.53820e+05  | -1.16164e+06 | -2.59441e+05 | 2.53823e+05  | -1.16165e+06 | -2.59444e+05 | 5.07643e+05  | -2.32330e+06 | -5.18885e+05 | 2.43406e+06 |
| 102   | 2.93034e+05  | -1.16587e+06 | -2.32976e+05 | 2.93037e+05  | -1.16588e+06 | -2.32978e+05 | 5.86071e+05  | -2.33175e+06 | -4.65955e+05 | 2.44901e+06 |
| 103   | 3.32360e+05  | -1.16763e+06 | -2.06284e+05 | 3.32362e+05  | -1.16764e+06 | -2.06286e+05 | 6.64722e+05  | -2.33528e+06 | -4.12570e+05 | 2.46284e+06 |
| 104   | 3.71686e+05  | -1.16687e+06 | -1.79489e+05 | 3.71689e+05  | -1.16688e+06 | -1.79491e+05 | 7.43375e+05  | -2.33375e+06 | -3.58980e+05 | 2.47545e+06 |
| 105   | 4.10895e+05  | -1.16352e+06 | -1.52719e+05 | 4.10898e+05  | -1.16353e+06 | -1.52721e+05 | 8.21793e+05  | -2.32705e+06 | -3.05440e+05 | 2.48672e+06 |
| 106   | 4.49861e+05  | -1.15754e+06 | -1.26108e+05 | 4.49864e+05  | -1.15755e+06 | -1.26109e+05 | 8.99725e+05  | -2.31509e+06 | -2.52217e+05 | 2.49655e+06 |
| 107   | 4.88452e+05  | -1.14890e+06 | -9.97899e+04 | 4.88456e+05  | -1.14891e+06 | -9.97907e+04 | 9.76908e+05  | -2.29781e+06 | -1.99581e+05 | 2.50482e+06 |
| 108   | 5.26530e+05  | -1.13758e+06 | -7.39038e+04 | 5.26534e+05  | -1.13759e+06 | -7.39044e+04 | 1.05306e+06  | -2.27516e+06 | -1.47808e+05 | 2.51141e+06 |
| 109   | 5.63949e+05  | -1.12357e+06 | -4.85889e+04 | 5.63953e+05  | -1.12358e+06 | -4.85893e+04 | 1.12790e+06  | -2.24715e+06 | -9.71782e+04 | 2.51620e+06 |
| 110   | 6.00561e+05  | -1.10689e+06 | -2.39846e+04 | 6.00565e+05  | -1.10690e+06 | -2.39848e+04 | 1.20113e+06  | -2.21378e+06 | -4.79694e+04 | 2.51910e+06 |
| 111   | 6.36210e+05  | -1.08756e+06 | -2.29318e+02 | 6.36215e+05  | -1.08756e+06 | -2.29319e+02 | 1.27243e+06  | -2.17512e+06 | -4.58637e+02 | 2.51996e+06 |
| 112   | 6.70739e+05  | -1.06562e+06 | 2.25409e+04  | 6.70744e+05  | -1.06563e+06 | 2.25411e+04  | 1.34148e+06  | -2.13124e+06 | 4.50820e+04  | 2.51869e+06 |
| 113   | 7.03985e+05  | -1.04113e+06 | 4.41937e+04  | 7.03991e+05  | -1.04114e+06 | 4.41941e+04  | 1.40798e+06  | -2.08227e+06 | 8.83878e+04  | 2.51517e+06 |
| 114   | 7.35786e+05  | -1.01418e+06 | 6.46016e+04  | 7.35791e+05  | -1.01419e+06 | 6.46021e+04  | 1.47158e+06  | -2.02836e+06 | 1.29204e+05  | 2.50928e+06 |
| 115   | 7.65975e+05  | -9.84844e+05 | 8.36436e+04  | 7.65981e+05  | -9.84852e+05 | 8.36442e+04  | 1.53196e+06  | -1.96970e+06 | 1.67288e+05  | 2.50092e+06 |
| 116   | 7.94389e+05  | -9.53245e+05 | 1.01206e+05  | 7.94396e+05  | -9.53252e+05 | 1.01207e+05  | 1.58879e+06  | -1.90650e+06 | 2.02414e+05  | 2.48997e+06 |
| 117   | 8.20864e+05  | -9.19507e+05 | 1.17186e+05  | 8.20870e+05  | -9.19514e+05 | 1.17187e+05  | 1.64173e+06  | -1.83902e+06 | 2.34373e+05  | 2.47633e+06 |
| 118   | 8.45236e+05  | -8.83775e+05 | 1.31489e+05  | 8.45243e+05  | -8.83783e+05 | 1.31490e+05  | 1.69048e+06  | -1.76756e+06 | 2.62979e+05  | 2.45991e+06 |
| 119   | 8.67347e+05  | -8.46211e+05 | 1.44034e+05  | 8.67355e+05  | -8.46219e+05 | 1.44035e+05  | 1.73470e+06  | -1.69243e+06 | 2.88069e+05  | 2.44059e+06 |
| 120   | 8.87043e+05  | -8.06992e+05 | 1.54753e+05  | 8.87051e+05  | -8.06999e+05 | 1.54754e+05  | 1.77409e+06  | -1.61399e+06 | 3.09506e+05  | 2.41830e+06 |
| 121   | 9.04175e+05  | -7.66307e+05 | 1.63591e+05  | 9.04184e+05  | -7.66314e+05 | 1.63592e+05  | 1.80836e+06  | -1.53262e+06 | 3.27183e+05  | 2.39294e+06 |
| 122   | 9.18601e+05  | -7.24363e+05 | 1.70511e+05  | 9.18611e+05  | -7.24370e+05 | 1.70512e+05  | 1.83721e+06  | -1.44873e+06 | 3.41023e+05  | 2.36442e+06 |
| 123   | 9.30188e+05  | -6.81378e+05 | 1.75490e+05  | 9.30198e+05  | -6.81385e+05 | 1.75492e+05  | 1.86039e+06  | -1.36276e+06 | 3.50983e+05  | 2.33267e+06 |
| 124   | 9.38812e+05  | -6.37580e+05 | 1.78526e+05  | 9.38822e+05  | -6.37588e+05 | 1.78528e+05  | 1.87763e+06  | -1.27517e+06 | 3.57054e+05  | 2.29762e+06 |
| 125   | 9.44358e+05  | -5.93210e+05 | 1.79632e+05  | 9.44370e+05  | -5.93217e+05 | 1.79634e+05  | 1.88873e+06  | -1.18643e+06 | 3.59266e+05  | 2.25920e+06 |
| 126   | 9.46726e+05  | -5.48514e+05 | 1.78841e+05  | 9.46739e+05  | -5.48521e+05 | 1.78843e+05  | 1.89346e+06  | -1.09704e+06 | 3.57684e+05  | 2.21735e+06 |
| 127   | 9.45828e+05  | -5.03747e+05 | 1.76207e+05  | 9.45841e+05  | -5.03754e+05 | 1.76209e+05  | 1.89167e+06  | -1.00750e+06 | 3.52415e+05  | 2.17202e+06 |
| 128   | 9.41589e+05  | -4.59167e+05 | 1.71801e+05  | 9.41603e+05  | -4.59174e+05 | 1.71803e+05  | 1.88319e+06  | -9.18341e+05 | 3.43604e+05  | 2.12316e+06 |
| 129   | 9.33951e+05  | -4.15037e+05 | 1.65716e+05  | 9.33966e+05  | -4.15044e+05 | 1.65718e+05  | 1.86792e+06  | -8.30081e+05 | 3.31434e+05  | 2.07075e+06 |

| phase | F+ x         | F+ y         | F+ z         | F- x         | F- y         | F- z         | Fnet x       | Fnet y       | Fnet z       | Fnet        |
|-------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|
| 130   | 9.22872e+05  | -3.71619e+05 | 1.58064e+05  | 9.22887e+05  | -3.71625e+05 | 1.58066e+05  | 1.84576e+06  | -7.43245e+05 | 3.16130e+05  | 2.01474e+06 |
| 131   | 9.08326e+05  | -3.29175e+05 | 1.48977e+05  | 9.08343e+05  | -3.29181e+05 | 1.48979e+05  | 1.81667e+06  | -6.58356e+05 | 2.97956e+05  | 1.95512e+06 |
| 132   | 8.90308e+05  | -2.87963e+05 | 1.38605e+05  | 8.90325e+05  | -2.87969e+05 | 1.38608e+05  | 1.78063e+06  | -5.75932e+05 | 2.77213e+05  | 1.89188e+06 |
| 133   | 8.68829e+05  | -2.48237e+05 | 1.27118e+05  | 8.68848e+05  | -2.48242e+05 | 1.27120e+05  | 1.73768e+06  | -4.96479e+05 | 2.54238e+05  | 1.82501e+06 |
| 134   | 8.43922e+05  | -2.10241e+05 | 1.14701e+05  | 8.43941e+05  | -2.10246e+05 | 1.14704e+05  | 1.68786e+06  | -4.20487e+05 | 2.29405e+05  | 1.75451e+06 |
| 135   | 8.15639e+05  | -1.74213e+05 | 1.01559e+05  | 8.15659e+05  | -1.74218e+05 | 1.01561e+05  | 1.63130e+06  | -3.48431e+05 | 2.03120e+05  | 1.68042e+06 |
| 136   | 7.84053e+05  | -1.40377e+05 | 8.79082e+04  | 7.84074e+05  | -1.40381e+05 | 8.79105e+04  | 1.56813e+06  | -2.80758e+05 | 1.75819e+05  | 1.60273e+06 |
| 137   | 7.49257e+05  | -1.08944e+05 | 7.39809e+04  | 7.49278e+05  | -1.08947e+05 | 7.39830e+04  | 1.49854e+06  | -2.17891e+05 | 1.47964e+05  | 1.52151e+06 |
| 138   | 7.11366e+05  | -8.01098e+04 | 6.00204e+04  | 7.11388e+05  | -8.01123e+04 | 6.00223e+04  | 1.42275e+06  | -1.60222e+05 | 1.20043e+05  | 1.43677e+06 |
| 139   | 6.70515e+05  | -5.40525e+04 | 4.62804e+04  | 6.70538e+05  | -5.40543e+04 | 4.62820e+04  | 1.34105e+06  | -1.08107e+05 | 9.25624e+04  | 1.34858e+06 |
| 140   | 6.26861e+05  | -3.09318e+04 | 3.30225e+04  | 6.26884e+05  | -3.09329e+04 | 3.30237e+04  | 1.25374e+06  | -6.18647e+04 | 6.60462e+04  | 1.25701e+06 |
| 141   | 5.80581e+05  | -1.08866e+04 | 2.05142e+04  | 5.80604e+05  | -1.08870e+04 | 2.05150e+04  | 1.16118e+06  | -2.17735e+04 | 4.10292e+04  | 1.16211e+06 |
| 142   | 5.31871e+05  | 5.96593e+03  | 9.02663e+03  | 5.31895e+05  | 5.96620e+03  | 9.02704e+03  | 1.06377e+06  | 1.19321e+04  | 1.80537e+04  | 1.06399e+06 |
| 143   | 4.80948e+05  | 1.95316e+04  | -1.16770e+03 | 4.80973e+05  | 1.95326e+04  | -1.16776e+03 | 9.61921e+05  | 3.90641e+04  | -2.33546e+03 | 9.62717e+05 |
| 144   | 4.28048e+05  | 2.97403e+04  | -9.79766e+03 | 4.28072e+05  | 2.97420e+04  | -9.79822e+03 | 8.56120e+05  | 5.94824e+04  | -1.95959e+04 | 8.58407e+05 |
| 145   | 3.73422e+05  | 3.65470e+04  | -1.65959e+04 | 3.73446e+05  | 3.65494e+04  | -1.65970e+04 | 7.46868e+05  | 7.30964e+04  | -3.31928e+04 | 7.51170e+05 |
| 146   | 3.17339e+05  | 3.99320e+04  | -2.13014e+04 | 3.17364e+05  | 3.99350e+04  | -2.13030e+04 | 6.34703e+05  | 7.98670e+04  | -4.26044e+04 | 6.41125e+05 |
| 147   | 2.60085e+05  | 3.99013e+04  | -2.36622e+04 | 2.60109e+05  | 3.99050e+04  | -2.36644e+04 | 5.20195e+05  | 7.98063e+04  | -4.73266e+04 | 5.28405e+05 |
| 148   | 2.01958e+05  | 3.64872e+04  | -2.34380e+04 | 2.01982e+05  | 3.64915e+04  | -2.34407e+04 | 4.03940e+05  | 7.29788e+04  | -4.68787e+04 | 4.13148e+05 |
| 149   | 1.43268e+05  | 2.97479e+04  | -2.04026e+04 | 1.43292e+05  | 2.97527e+04  | -2.04059e+04 | 2.86560e+05  | 5.95006e+04  | -4.08084e+04 | 2.95503e+05 |
| 150   | 8.43374e+04  | 1.97668e+04  | -1.43465e+04 | 8.43604e+04  | 1.97722e+04  | -1.43504e+04 | 1.68698e+05  | 3.95391e+04  | -2.86970e+04 | 1.75630e+05 |
| 151   | 2.54959e+04  | 6.65300e+03  | -5.07960e+03 | 2.55183e+04  | 6.65885e+03  | -5.08407e+03 | 5.10142e+04  | 1.33118e+04  | -1.01637e+04 | 5.36931e+04 |
| 152   | -3.29189e+04 | -9.46067e+03 | 7.56699e+03  | -3.28972e+04 | -9.45442e+03 | 7.56199e+03  | -6.58161e+04 | -1.89151e+04 | 1.51290e+04  | 7.01315e+04 |
| 153   | -9.05647e+04 | -2.84179e+04 | 2.37384e+04  | -9.05437e+04 | -2.84113e+04 | 2.37329e+04  | -1.81108e+05 | -5.68292e+04 | 4.74712e+04  | 1.95661e+05 |
| 154   | -1.47096e+05 | -5.00404e+04 | 4.35539e+04  | -1.47076e+05 | -5.00335e+04 | 4.35479e+04  | -2.94172e+05 | -1.00074e+05 | 8.71018e+04  | 3.22706e+05 |
| 155   | -2.02168e+05 | -7.41292e+04 | 6.71054e+04  | -2.02148e+05 | -7.41221e+04 | 6.70990e+04  | -4.04316e+05 | -1.48251e+05 | 1.34204e+05  | 4.51067e+05 |
| 156   | -2.55435e+05 | -1.00467e+05 | 9.44554e+04  | -2.55417e+05 | -1.00459e+05 | 9.44486e+04  | -5.10852e+05 | -2.00926e+05 | 1.88904e+05  | 5.80539e+05 |
| 157   | -3.06558e+05 | -1.28818e+05 | 1.25636e+05  | -3.06540e+05 | -1.28811e+05 | 1.25629e+05  | -6.13098e+05 | -2.57629e+05 | 2.51265e+05  | 7.10912e+05 |
| 158   | -3.55201e+05 | -1.58933e+05 | 1.60647e+05  | -3.55185e+05 | -1.58926e+05 | 1.60640e+05  | -7.10386e+05 | -3.17859e+05 | 3.21287e+05  | 8.41966e+05 |
| 159   | -4.01040e+05 | -1.90549e+05 | 1.99456e+05  | -4.01025e+05 | -1.90542e+05 | 1.99449e+05  | -8.02065e+05 | -3.81091e+05 | 3.98905e+05  | 9.73480e+05 |
| 160   | -4.43760e+05 | -2.23392e+05 | 2.41997e+05  | -4.43745e+05 | -2.23385e+05 | 2.41990e+05  | -8.87505e+05 | -4.46777e+05 | 4.83987e+05  | 1.10522e+06 |
| 161   | -4.83059e+05 | -2.57181e+05 | 2.88171e+05  | -4.83045e+05 | -2.57174e+05 | 2.88163e+05  | -9.66104e+05 | -5.14356e+05 | 5.76334e+05  | 1.23696e+06 |
| 162   | -5.18651e+05 | -2.91628e+05 | 3.37843e+05  | -5.18638e+05 | -2.91621e+05 | 3.37835e+05  | -1.03729e+06 | -5.83250e+05 | 6.75678e+05  | 1.36846e+06 |
| 163   | -5.50268e+05 | -3.26443e+05 | 3.90847e+05  | -5.50256e+05 | -3.26436e+05 | 3.90839e+05  | -1.10052e+06 | -6.52878e+05 | 7.81686e+05  | 1.49948e+06 |
| 164   | -5.77662e+05 | -3.61332e+05 | 4.46983e+05  | -5.77651e+05 | -3.61326e+05 | 4.46974e+05  | -1.15531e+06 | -7.22658e+05 | 8.93957e+05  | 1.62977e+06 |
| 165   | -6.00605e+05 | -3.96008e+05 | 5.06018e+05  | -6.00595e+05 | -3.96002e+05 | 5.06010e+05  | -1.20120e+06 | -7.92010e+05 | 1.01203e+06  | 1.75908e+06 |
| 166   | -6.18896e+05 | -4.30185e+05 | 5.67690e+05  | -6.18887e+05 | -4.30179e+05 | 5.67682e+05  | -1.23778e+06 | -8.60364e+05 | 1.13537e+06  | 1.88717e+06 |
| 167   | -6.32359e+05 | -4.63585e+05 | 6.31706e+05  | -6.32350e+05 | -4.63579e+05 | 6.31698e+05  | -1.26471e+06 | -9.27164e+05 | 1.26340e+06  | 2.01378e+06 |
| 168   | -6.40842e+05 | -4.95940e+05 | 6.97746e+05  | -6.40835e+05 | -4.95934e+05 | 6.97737e+05  | -1.28168e+06 | -9.91874e+05 | 1.39548e+06  | 2.13866e+06 |
| 169   | -6.44227e+05 | -5.26992e+05 | 7.65464e+05  | -6.44221e+05 | -5.26986e+05 | 7.65455e+05  | -1.28845e+06 | -1.05398e+06 | 1.53092e+06  | 2.26157e+06 |
| 170   | -6.42424e+05 | -5.56500e+05 | 8.34491e+05  | -6.42418e+05 | -5.56494e+05 | 8.34483e+05  | -1.28484e+06 | -1.11299e+06 | 1.66897e+06  | 2.38224e+06 |
| 171   | -6.35373e+05 | -5.84237e+05 | 9.04437e+05  | -6.35368e+05 | -5.84232e+05 | 9.04429e+05  | -1.27074e+06 | -1.16847e+06 | 1.80887e+06  | 2.50042e+06 |
| 172   | -6.23050e+05 | -6.09995e+05 | 9.74896e+05  | -6.23045e+05 | -6.09990e+05 | 9.74888e+05  | -1.24610e+06 | -1.21999e+06 | 1.94978e+06  | 2.61587e+06 |
| 173   | -6.05462e+05 | -6.33587e+05 | 1.04544e+06  | -6.05457e+05 | -6.33583e+05 | 1.04544e+06  | -1.21092e+06 | -1.26717e+06 | 2.09088e+06  | 2.72834e+06 |
| 174   | -5.82649e+05 | -6.54847e+05 | 1.11565e+06  | -5.82644e+05 | -6.54842e+05 | 1.11564e+06  | -1.16529e+06 | -1.30969e+06 | 2.23128e+06  | 2.83757e+06 |

| phase | F+ x         | F+ y         | F+ z        | F- x         | F- y         | F- z        | Fnet x       | Fnet y       | Fnet z      | Fnet        |
|-------|--------------|--------------|-------------|--------------|--------------|-------------|--------------|--------------|-------------|-------------|
| 175   | -5.54687e+05 | -6.73631e+05 | 1.18506e+06 | -5.54683e+05 | -6.73626e+05 | 1.18505e+06 | -1.10937e+06 | -1.34726e+06 | 2.37012e+06 | 2.94334e+06 |
| 176   | -5.21687e+05 | -6.89821e+05 | 1.25324e+06 | -5.21683e+05 | -6.89816e+05 | 1.25324e+06 | -1.04337e+06 | -1.37964e+06 | 2.50648e+06 | 3.04540e+06 |
| 177   | -4.83791e+05 | -7.03322e+05 | 1.31974e+06 | -4.83788e+05 | -7.03317e+05 | 1.31974e+06 | -9.67579e+05 | -1.40664e+06 | 2.63948e+06 | 3.14352e+06 |
| 178   | -4.41178e+05 | -7.14066e+05 | 1.38412e+06 | -4.41175e+05 | -7.14061e+05 | 1.38411e+06 | -8.82353e+05 | -1.42813e+06 | 2.76822e+06 | 3.23746e+06 |
| 179   | -3.94057e+05 | -7.22012e+05 | 1.44592e+06 | -3.94055e+05 | -7.22007e+05 | 1.44591e+06 | -7.88112e+05 | -1.44402e+06 | 2.89183e+06 | 3.32701e+06 |
| 180   | -3.42672e+05 | -7.27144e+05 | 1.50473e+06 | -3.42670e+05 | -7.27139e+05 | 1.50472e+06 | -6.85342e+05 | -1.45428e+06 | 3.00945e+06 | 3.41195e+06 |
| 181   | -2.87296e+05 | -7.29471e+05 | 1.56013e+06 | -2.87294e+05 | -7.29466e+05 | 1.56012e+06 | -5.74590e+05 | -1.45894e+06 | 3.12025e+06 | 3.49207e+06 |
| 182   | -2.28233e+05 | -7.29031e+05 | 1.61172e+06 | -2.28231e+05 | -7.29025e+05 | 1.61170e+06 | -4.56464e+05 | -1.45806e+06 | 3.22342e+06 | 3.56717e+06 |
| 183   | -1.65814e+05 | -7.25883e+05 | 1.65911e+06 | -1.65813e+05 | -7.25877e+05 | 1.65910e+06 | -3.31626e+05 | -1.45176e+06 | 3.31822e+06 | 3.63705e+06 |
| 184   | -1.00397e+05 | -7.20112e+05 | 1.70197e+06 | -1.00396e+05 | -7.20107e+05 | 1.70196e+06 | -2.00793e+05 | -1.44022e+06 | 3.40393e+06 | 3.70153e+06 |
| 185   | -3.23634e+04 | -7.11827e+05 | 1.73997e+06 | -3.23631e+04 | -7.11821e+05 | 1.73995e+06 | -6.47264e+04 | -1.42365e+06 | 3.47992e+06 | 3.76043e+06 |
| 186   | 3.78822e+04  | -7.01156e+05 | 1.77280e+06 | 3.78818e+04  | -7.01150e+05 | 1.77278e+06 | 7.57640e+04  | -1.40231e+06 | 3.54558e+06 | 3.81357e+06 |
| 187   | 1.09916e+05  | -6.88248e+05 | 1.80021e+06 | 1.09915e+05  | -6.88242e+05 | 1.80019e+06 | 2.19832e+05  | -1.37649e+06 | 3.60040e+06 | 3.86082e+06 |
| 188   | 1.83299e+05  | -6.73269e+05 | 1.82197e+06 | 1.83297e+05  | -6.73263e+05 | 1.82195e+06 | 3.66596e+05  | -1.34653e+06 | 3.64392e+06 | 3.90201e+06 |
| 189   | 2.57575e+05  | -6.56402e+05 | 1.83789e+06 | 2.57573e+05  | -6.56396e+05 | 1.83788e+06 | 5.15148e+05  | -1.31280e+06 | 3.67577e+06 | 3.93702e+06 |
| 190   | 3.32279e+05  | -6.37843e+05 | 1.84784e+06 | 3.32276e+05  | -6.37837e+05 | 1.84782e+06 | 6.64556e+05  | -1.27568e+06 | 3.69566e+06 | 3.96571e+06 |
| 191   | 4.06938e+05  | -6.17799e+05 | 1.85169e+06 | 4.06933e+05  | -6.17793e+05 | 1.85167e+06 | 8.13871e+05  | -1.23559e+06 | 3.70337e+06 | 3.98798e+06 |
| 192   | 4.81070e+05  | -5.96486e+05 | 1.84940e+06 | 4.81065e+05  | -5.96479e+05 | 1.84938e+06 | 9.62136e+05  | -1.19297e+06 | 3.69877e+06 | 4.00372e+06 |
| 193   | 5.54197e+05  | -5.74125e+05 | 1.84093e+06 | 5.54191e+05  | -5.74118e+05 | 1.84091e+06 | 1.10839e+06  | -1.14824e+06 | 3.68185e+06 | 4.01285e+06 |
| 194   | 6.25837e+05  | -5.50942e+05 | 1.82633e+06 | 6.25830e+05  | -5.50935e+05 | 1.82631e+06 | 1.25167e+06  | -1.10188e+06 | 3.65264e+06 | 4.01529e+06 |
| 195   | 6.95516e+05  | -5.27163e+05 | 1.80566e+06 | 6.95508e+05  | -5.27157e+05 | 1.80564e+06 | 1.39102e+06  | -1.05432e+06 | 3.61129e+06 | 4.01098e+06 |
| 196   | 7.62767e+05  | -5.03013e+05 | 1.77903e+06 | 7.62757e+05  | -5.03006e+05 | 1.77901e+06 | 1.52552e+06  | -1.00602e+06 | 3.55804e+06 | 3.99987e+06 |
| 197   | 8.27134e+05  | -4.78710e+05 | 1.74662e+06 | 8.27123e+05  | -4.78704e+05 | 1.74659e+06 | 1.65426e+06  | -9.57414e+05 | 3.49321e+06 | 3.98192e+06 |
| 198   | 8.88175e+05  | -4.54468e+05 | 1.70861e+06 | 8.88163e+05  | -4.54462e+05 | 1.70859e+06 | 1.77634e+06  | -9.08930e+05 | 3.41720e+06 | 3.95712e+06 |
| 199   | 9.45467e+05  | -4.30489e+05 | 1.66527e+06 | 9.45453e+05  | -4.30483e+05 | 1.66524e+06 | 1.89092e+06  | -8.60973e+05 | 3.33051e+06 | 3.92545e+06 |
| 200   | 9.98607e+05  | -4.06964e+05 | 1.61686e+06 | 9.98591e+05  | -4.06958e+05 | 1.61684e+06 | 1.99720e+06  | -8.13922e+05 | 3.23370e+06 | 3.88691e+06 |
| 201   | 1.04721e+06  | -3.84069e+05 | 1.56373e+06 | 1.04720e+06  | -3.84063e+05 | 1.56371e+06 | 2.09441e+06  | -7.68132e+05 | 3.12744e+06 | 3.84154e+06 |
| 202   | 1.09094e+06  | -3.61962e+05 | 1.50623e+06 | 1.09092e+06  | -3.61956e+05 | 1.50620e+06 | 2.18186e+06  | -7.23919e+05 | 3.01243e+06 | 3.78937e+06 |
| 203   | 1.12945e+06  | -3.40786e+05 | 1.44475e+06 | 1.12944e+06  | -3.40780e+05 | 1.44472e+06 | 2.25889e+06  | -6.81565e+05 | 2.88947e+06 | 3.73044e+06 |
| 204   | 1.16247e+06  | -3.20659e+05 | 1.37971e+06 | 1.16245e+06  | -3.20653e+05 | 1.37969e+06 | 2.32492e+06  | -6.41312e+05 | 2.75940e+06 | 3.66481e+06 |
| 205   | 1.18973e+06  | -3.01680e+05 | 1.31157e+06 | 1.18971e+06  | -3.01675e+05 | 1.31155e+06 | 2.37944e+06  | -6.03355e+05 | 2.62312e+06 | 3.59257e+06 |
| 206   | 1.21101e+06  | -2.83925e+05 | 1.24080e+06 | 1.21099e+06  | -2.83920e+05 | 1.24078e+06 | 2.42200e+06  | -5.67845e+05 | 2.48158e+06 | 3.51380e+06 |
| 207   | 1.22613e+06  | -2.67444e+05 | 1.16789e+06 | 1.22611e+06  | -2.67439e+05 | 1.16787e+06 | 2.45224e+06  | -5.34883e+05 | 2.33577e+06 | 3.42861e+06 |
| 208   | 1.23495e+06  | -2.52264e+05 | 1.09336e+06 | 1.23492e+06  | -2.52259e+05 | 1.09334e+06 | 2.46987e+06  | -5.04522e+05 | 2.18669e+06 | 3.33713e+06 |
| 209   | 1.23735e+06  | -2.38384e+05 | 1.01771e+06 | 1.23733e+06  | -2.38379e+05 | 1.01769e+06 | 2.47468e+06  | -4.76763e+05 | 2.03540e+06 | 3.23947e+06 |
| 210   | 1.23329e+06  | -2.25779e+05 | 9.41482e+05 | 1.23326e+06  | -2.25775e+05 | 9.41462e+05 | 2.46655e+06  | -4.51554e+05 | 1.88294e+06 | 3.13580e+06 |
| 211   | 1.22273e+06  | -2.14400e+05 | 8.65201e+05 | 1.22271e+06  | -2.14395e+05 | 8.65182e+05 | 2.44544e+06  | -4.28795e+05 | 1.73038e+06 | 3.02626e+06 |
| 212   | 1.20572e+06  | -2.04169e+05 | 7.89396e+05 | 1.20569e+06  | -2.04164e+05 | 7.89379e+05 | 2.41141e+06  | -4.08333e+05 | 1.57878e+06 | 2.91104e+06 |
| 213   | 1.18230e+06  | -1.94986e+05 | 7.14592e+05 | 1.18228e+06  | -1.94982e+05 | 7.14576e+05 | 2.36458e+06  | -3.89968e+05 | 1.42917e+06 | 2.79031e+06 |
| 214   | 1.15261e+06  | -1.86729e+05 | 6.41300e+05 | 1.15258e+06  | -1.86724e+05 | 6.41285e+05 | 2.30519e+06  | -3.73453e+05 | 1.28259e+06 | 2.66428e+06 |
| 215   | 1.11678e+06  | -1.79250e+05 | 5.70022e+05 | 1.11676e+06  | -1.79246e+05 | 5.70008e+05 | 2.23354e+06  | -3.58496e+05 | 1.14003e+06 | 2.53316e+06 |
| 216   | 1.07502e+06  | -1.72384e+05 | 5.01237e+05 | 1.07499e+06  | -1.72380e+05 | 5.01225e+05 | 2.15002e+06  | -3.44764e+05 | 1.00246e+06 | 2.39716e+06 |
| 217   | 1.02756e+06  | -1.65945e+05 | 4.35405e+05 | 1.02754e+06  | -1.65941e+05 | 4.35394e+05 | 2.05510e+06  | -3.31886e+05 | 8.70800e+05 | 2.25652e+06 |
| 218   | 9.74682e+05  | -1.59732e+05 | 3.72960e+05 | 9.74657e+05  | -1.59728e+05 | 3.72950e+05 | 1.94934e+06  | -3.19460e+05 | 7.45910e+05 | 2.11148e+06 |
| 219   | 9.16694e+05  | -1.53528e+05 | 3.14306e+05 | 9.16668e+05  | -1.53524e+05 | 3.14297e+05 | 1.83336e+06  | -3.07052e+05 | 6.28603e+05 | 1.96230e+06 |

| phase | F+ x         | F+ y         | F+ z         | F- x         | F- y         | F- z         | Fnet x       | Fnet y       | Fnet z       | Fnet        |
|-------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|
| 220   | 8.53946e+05  | -1.47105e+05 | 2.59814e+05  | 8.53922e+05  | -1.47101e+05 | 2.59806e+05  | 1.70787e+06  | -2.94206e+05 | 5.19620e+05  | 1.80925e+06 |
| 221   | 7.86824e+05  | -1.40225e+05 | 2.09820e+05  | 7.86800e+05  | -1.40221e+05 | 2.09814e+05  | 1.57362e+06  | -2.80447e+05 | 4.19634e+05  | 1.65258e+06 |
| 222   | 7.15740e+05  | -1.32645e+05 | 1.64623e+05  | 7.15717e+05  | -1.32641e+05 | 1.64618e+05  | 1.43146e+06  | -2.65286e+05 | 3.29241e+05  | 1.49260e+06 |
| 223   | 6.41137e+05  | -1.24116e+05 | 1.24480e+05  | 6.41115e+05  | -1.24112e+05 | 1.24476e+05  | 1.28225e+06  | -2.48228e+05 | 2.48956e+05  | 1.32957e+06 |
| 224   | 5.63484e+05  | -1.14391e+05 | 8.96068e+04  | 5.63463e+05  | -1.14386e+05 | 8.96035e+04  | 1.12695e+06  | -2.28777e+05 | 1.79210e+05  | 1.16382e+06 |
| 225   | 4.83271e+05  | -1.03222e+05 | 6.01741e+04  | 4.83251e+05  | -1.03218e+05 | 6.01717e+04  | 9.66521e+05  | -2.06440e+05 | 1.20346e+05  | 9.95623e+05 |
| 226   | 4.01006e+05  | -9.03707e+04 | 3.63081e+04  | 4.00987e+05  | -9.03665e+04 | 3.63064e+04  | 8.01994e+05  | -1.80737e+05 | 7.26145e+04  | 8.25308e+05 |
| 227   | 3.17216e+05  | -7.56042e+04 | 1.80888e+04  | 3.17198e+05  | -7.56000e+04 | 1.80878e+04  | 6.34414e+05  | -1.51204e+05 | 3.61766e+04  | 6.53186e+05 |
| 228   | 2.32436e+05  | -5.87025e+04 | 5.54978e+03  | 2.32419e+05  | -5.86984e+04 | 5.54939e+03  | 4.64855e+05  | -1.17401e+05 | 1.10992e+04  | 4.79580e+05 |
| 229   | 1.47212e+05  | -3.94600e+04 | -1.32164e+03 | 1.47197e+05  | -3.94559e+04 | -1.32151e+03 | 2.94408e+05  | -7.89159e+04 | -2.64315e+03 | 3.04813e+05 |
| 230   | 6.20933e+04  | -1.76886e+04 | -2.58423e+03 | 6.20790e+04  | -1.76845e+04 | -2.58364e+03 | 1.24172e+05  | -3.53732e+04 | -5.16787e+03 | 1.29216e+05 |
| 231   | -2.23697e+04 | 6.77975e+03  | 1.65814e+03  | -2.23830e+04 | 6.78377e+03  | 1.65913e+03  | -4.47527e+04 | 1.35635e+04  | 3.31727e+03  | 4.68804e+04 |
| 232   | -1.05630e+05 | 3.40903e+04  | 1.12577e+04  | -1.05642e+05 | 3.40943e+04  | 1.12590e+04  | -2.11272e+05 | 6.81847e+04  | 2.25166e+04  | 2.23142e+05 |
| 233   | -1.87148e+05 | 6.43634e+04  | 2.60242e+04  | -1.87159e+05 | 6.43673e+04  | 2.60258e+04  | -3.74307e+05 | 1.28731e+05  | 5.20499e+04  | 3.99232e+05 |
| 234   | -2.66393e+05 | 9.76918e+04  | 4.57271e+04  | -2.66404e+05 | 9.76957e+04  | 4.57289e+04  | -5.32797e+05 | 1.95388e+05  | 9.14560e+04  | 5.74816e+05 |
| 235   | -3.42852e+05 | 1.34140e+05  | 7.00979e+04  | -3.42862e+05 | 1.34143e+05  | 7.00999e+04  | -6.85713e+05 | 2.68283e+05  | 1.40198e+05  | 7.49556e+05 |
| 236   | -4.16027e+05 | 1.73740e+05  | 9.88323e+04  | -4.16036e+05 | 1.73744e+05  | 9.88345e+04  | -8.32063e+05 | 3.47484e+05  | 1.97667e+05  | 9.23118e+05 |
| 237   | -4.85444e+05 | 2.16494e+05  | 1.31594e+05  | -4.85452e+05 | 2.16498e+05  | 1.31596e+05  | -9.70896e+05 | 4.32991e+05  | 2.63190e+05  | 1.09517e+06 |
| 238   | -5.50652e+05 | 2.62369e+05  | 1.68017e+05  | -5.50660e+05 | 2.62373e+05  | 1.68019e+05  | -1.10131e+06 | 5.24743e+05  | 3.36036e+05  | 1.26537e+06 |
| 239   | -6.11230e+05 | 3.11301e+05  | 2.07709e+05  | -6.11238e+05 | 3.11305e+05  | 2.07711e+05  | -1.22247e+06 | 6.22606e+05  | 4.15420e+05  | 1.43340e+06 |
| 240   | -6.66787e+05 | 3.63189e+05  | 2.50256e+05  | -6.66795e+05 | 3.63193e+05  | 2.50259e+05  | -1.33358e+06 | 7.26382e+05  | 5.00515e+05  | 1.59893e+06 |
| 241   | -7.16967e+05 | 4.17899e+05  | 2.95226e+05  | -7.16974e+05 | 4.17903e+05  | 2.95229e+05  | -1.43394e+06 | 8.35802e+05  | 5.90456e+05  | 1.76164e+06 |
| 242   | -7.61450e+05 | 4.75264e+05  | 3.42172e+05  | -7.61457e+05 | 4.75268e+05  | 3.42175e+05  | -1.52291e+06 | 9.50531e+05  | 6.84348e+05  | 1.92122e+06 |
| 243   | -7.99954e+05 | 5.35082e+05  | 3.90637e+05  | -7.99961e+05 | 5.35086e+05  | 3.90640e+05  | -1.59991e+06 | 1.07017e+06  | 7.81277e+05  | 2.07735e+06 |
| 244   | -8.32239e+05 | 5.97121e+05  | 4.40156e+05  | -8.32246e+05 | 5.97125e+05  | 4.40159e+05  | -1.66449e+06 | 1.19425e+06  | 8.80315e+05  | 2.22973e+06 |
| 245   | -8.58108e+05 | 6.61116e+05  | 4.90264e+05  | -8.58115e+05 | 6.61122e+05  | 4.90268e+05  | -1.71622e+06 | 1.32224e+06  | 9.80531e+05  | 2.37806e+06 |
| 246   | -8.77406e+05 | 7.26776e+05  | 5.40496e+05  | -8.77413e+05 | 7.26782e+05  | 5.40501e+05  | -1.75482e+06 | 1.45356e+06  | 1.08100e+06  | 2.52206e+06 |
| 247   | -8.90026e+05 | 7.93780e+05  | 5.90396e+05  | -8.90032e+05 | 7.93786e+05  | 5.90400e+05  | -1.78006e+06 | 1.58757e+06  | 1.18080e+06  | 2.66144e+06 |
| 248   | -8.95903e+05 | 8.61784e+05  | 6.39514e+05  | -8.95910e+05 | 8.61791e+05  | 6.39519e+05  | -1.79181e+06 | 1.72358e+06  | 1.27903e+06  | 2.79593e+06 |
| 249   | -8.95023e+05 | 9.30421e+05  | 6.87417e+05  | -8.95029e+05 | 9.30428e+05  | 6.87422e+05  | -1.79005e+06 | 1.86085e+06  | 1.37484e+06  | 2.92527e+06 |
| 250   | -8.87414e+05 | 9.99305e+05  | 7.33687e+05  | -8.87421e+05 | 9.99313e+05  | 7.33692e+05  | -1.77483e+06 | 1.99862e+06  | 1.46738e+06  | 3.04922e+06 |
| 251   | -8.73153e+05 | 1.06803e+06  | 7.77928e+05  | -8.73160e+05 | 1.06804e+06  | 7.77934e+05  | -1.74631e+06 | 2.13608e+06  | 1.55586e+06  | 3.16751e+06 |
| 252   | -8.52362e+05 | 1.13619e+06  | 8.19770e+05  | -8.52369e+05 | 1.13620e+06  | 8.19776e+05  | -1.70473e+06 | 2.27240e+06  | 1.63955e+06  | 3.27994e+06 |
| 253   | -8.25208e+05 | 1.20336e+06  | 8.58866e+05  | -8.25215e+05 | 1.20337e+06  | 8.58873e+05  | -1.65042e+06 | 2.40673e+06  | 1.71774e+06  | 3.38628e+06 |
| 254   | -7.91902e+05 | 1.26910e+06  | 8.94903e+05  | -7.91908e+05 | 1.26911e+06  | 8.94911e+05  | -1.58381e+06 | 2.53821e+06  | 1.78981e+06  | 3.48632e+06 |
| 255   | -7.52695e+05 | 1.33298e+06  | 9.27598e+05  | -7.52702e+05 | 1.33300e+06  | 9.27606e+05  | -1.50540e+06 | 2.66598e+06  | 1.85520e+06  | 3.57987e+06 |
| 256   | -7.07881e+05 | 1.39458e+06  | 9.56701e+05  | -7.07888e+05 | 1.39459e+06  | 9.56710e+05  | -1.41577e+06 | 2.78917e+06  | 1.91341e+06  | 3.66675e+06 |
| 257   | -6.57792e+05 | 1.45347e+06  | 9.82001e+05  | -6.57798e+05 | 1.45348e+06  | 9.82010e+05  | -1.31559e+06 | 2.90695e+06  | 1.96401e+06  | 3.74679e+06 |
| 258   | -6.02793e+05 | 1.50923e+06  | 1.00332e+06  | -6.02799e+05 | 1.50924e+06  | 1.00333e+06  | -1.20559e+06 | 3.01846e+06  | 2.00665e+06  | 3.81985e+06 |
| 259   | -5.43287e+05 | 1.56146e+06  | 1.02053e+06  | -5.43292e+05 | 1.56147e+06  | 1.02054e+06  | -1.08658e+06 | 3.12293e+06  | 2.04107e+06  | 3.88578e+06 |
| 260   | -4.79704e+05 | 1.60977e+06  | 1.03352e+06  | -4.79708e+05 | 1.60979e+06  | 1.03353e+06  | -9.59412e+05 | 3.21956e+06  | 2.06705e+06  | 3.94446e+06 |
| 261   | -4.12502e+05 | 1.65381e+06  | 1.04224e+06  | -4.12506e+05 | 1.65383e+06  | 1.04225e+06  | -8.25007e+05 | 3.30764e+06  | 2.08449e+06  | 3.99578e+06 |
| 262   | -3.42164e+05 | 1.69323e+06  | 1.04667e+06  | -3.42167e+05 | 1.69325e+06  | 1.04668e+06  | -6.84331e+05 | 3.38649e+06  | 2.09335e+06  | 4.03964e+06 |
| 263   | -2.69193e+05 | 1.72773e+06  | 1.04682e+06  | -2.69196e+05 | 1.72775e+06  | 1.04683e+06  | -5.38388e+05 | 3.45547e+06  | 2.09365e+06  | 4.07597e+06 |
| 264   | -1.94110e+05 | 1.75701e+06  | 1.04276e+06  | -1.94112e+05 | 1.75703e+06  | 1.04277e+06  | -3.88222e+05 | 3.51404e+06  | 2.08552e+06  | 4.10471e+06 |

| phase | F+ x         | F+ y         | F+ z         | F- x         | F- y         | F- z         | Fnet x       | Fnet y       | Fnet z       | Fnet        |
|-------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|
| 265   | -1.17449e+05 | 1.78084e+06  | 1.03456e+06  | -1.17450e+05 | 1.78086e+06  | 1.03458e+06  | -2.34900e+05 | 3.56170e+06  | 2.06914e+06  | 4.12580e+06 |
| 266   | -3.97547e+04 | 1.79900e+06  | 1.02237e+06  | -3.97552e+04 | 1.79902e+06  | 1.02238e+06  | -7.95099e+04 | 3.59801e+06  | 2.04476e+06  | 4.13921e+06 |
| 267   | 3.84234e+04  | 1.81131e+06  | 1.00634e+06  | 3.84238e+04  | 1.81133e+06  | 1.00635e+06  | 7.68473e+04  | 3.62264e+06  | 2.01270e+06  | 4.14492e+06 |
| 268   | 1.16533e+05  | 1.81766e+06  | 9.86661e+05  | 1.16534e+05  | 1.81767e+06  | 9.86671e+05  | 2.33067e+05  | 3.63533e+06  | 1.97333e+06  | 4.14294e+06 |
| 269   | 1.94023e+05  | 1.81794e+06  | 9.63547e+05  | 1.94025e+05  | 1.81795e+06  | 9.63558e+05  | 3.88049e+05  | 3.63589e+06  | 1.92711e+06  | 4.13328e+06 |
| 270   | 2.70349e+05  | 1.81211e+06  | 9.37246e+05  | 2.70352e+05  | 1.81213e+06  | 9.37256e+05  | 5.40701e+05  | 3.62423e+06  | 1.87450e+06  | 4.11597e+06 |
| 271   | 3.44975e+05  | 1.80017e+06  | 9.08022e+05  | 3.44978e+05  | 1.80019e+06  | 9.08031e+05  | 6.89953e+05  | 3.60035e+06  | 1.81605e+06  | 4.09104e+06 |
| 272   | 4.17376e+05  | 1.78216e+06  | 8.76160e+05  | 4.17381e+05  | 1.78218e+06  | 8.76170e+05  | 8.34757e+05  | 3.56434e+06  | 1.75233e+06  | 4.05857e+06 |
| 273   | 4.87048e+05  | 1.75818e+06  | 8.41963e+05  | 4.87053e+05  | 1.75820e+06  | 8.41972e+05  | 9.74101e+05  | 3.51637e+06  | 1.68394e+06  | 4.01863e+06 |
| 274   | 5.53502e+05  | 1.72835e+06  | 8.05744e+05  | 5.53508e+05  | 1.72837e+06  | 8.05753e+05  | 1.10701e+06  | 3.45672e+06  | 1.61150e+06  | 3.97131e+06 |
| 275   | 6.16275e+05  | 1.69285e+06  | 7.67827e+05  | 6.16281e+05  | 1.69287e+06  | 7.67835e+05  | 1.23256e+06  | 3.38573e+06  | 1.53566e+06  | 3.91671e+06 |
| 276   | 6.74929e+05  | 1.65191e+06  | 7.28540e+05  | 6.74936e+05  | 1.65193e+06  | 7.28548e+05  | 1.34987e+06  | 3.30385e+06  | 1.45709e+06  | 3.85495e+06 |
| 277   | 7.29057e+05  | 1.60579e+06  | 6.88215e+05  | 7.29064e+05  | 1.60581e+06  | 6.88222e+05  | 1.45812e+06  | 3.21161e+06  | 1.37644e+06  | 3.78617e+06 |
| 278   | 7.78281e+05  | 1.55480e+06  | 6.47180e+05  | 7.78289e+05  | 1.55481e+06  | 6.47187e+05  | 1.55657e+06  | 3.10961e+06  | 1.29437e+06  | 3.71052e+06 |
| 279   | 8.22259e+05  | 1.49926e+06  | 6.05763e+05  | 8.22267e+05  | 1.49928e+06  | 6.05769e+05  | 1.64453e+06  | 2.99854e+06  | 1.21153e+06  | 3.62815e+06 |
| 280   | 8.60686e+05  | 1.43957e+06  | 5.64278e+05  | 8.60694e+05  | 1.43958e+06  | 5.64284e+05  | 1.72138e+06  | 2.87915e+06  | 1.12856e+06  | 3.53925e+06 |
| 281   | 8.93293e+05  | 1.37612e+06  | 5.23033e+05  | 8.93301e+05  | 1.37613e+06  | 5.23038e+05  | 1.78659e+06  | 2.75225e+06  | 1.04607e+06  | 3.44399e+06 |
| 282   | 9.19853e+05  | 1.30936e+06  | 4.82318e+05  | 9.19861e+05  | 1.30937e+06  | 4.82322e+05  | 1.83971e+06  | 2.61874e+06  | 9.64640e+05  | 3.34258e+06 |
| 283   | 9.40179e+05  | 1.23976e+06  | 4.42407e+05  | 9.40188e+05  | 1.23977e+06  | 4.42411e+05  | 1.88037e+06  | 2.47953e+06  | 8.84818e+05  | 3.23523e+06 |
| 284   | 9.54128e+05  | 1.16779e+06  | 4.03555e+05  | 9.54137e+05  | 1.16780e+06  | 4.03558e+05  | 1.90826e+06  | 2.33559e+06  | 8.07113e+05  | 3.12216e+06 |
| 285   | 9.61599e+05  | 1.09397e+06  | 3.65993e+05  | 9.61607e+05  | 1.09398e+06  | 3.65996e+05  | 1.92321e+06  | 2.18795e+06  | 7.31990e+05  | 3.00361e+06 |
| 286   | 9.62534e+05  | 1.01882e+06  | 3.29930e+05  | 9.62542e+05  | 1.01882e+06  | 3.29933e+05  | 1.92508e+06  | 2.03764e+06  | 6.59863e+05  | 2.87981e+06 |
| 287   | 9.56918e+05  | 9.42856e+05  | 2.95546e+05  | 9.56926e+05  | 9.42863e+05  | 2.95549e+05  | 1.91384e+06  | 1.88572e+06  | 5.91095e+05  | 2.75102e+06 |
| 288   | 9.44783e+05  | 8.66628e+05  | 2.62995e+05  | 9.44790e+05  | 8.66634e+05  | 2.62998e+05  | 1.88957e+06  | 1.73326e+06  | 5.25993e+05  | 2.61751e+06 |
| 289   | 9.26199e+05  | 7.90670e+05  | 2.32402e+05  | 9.26206e+05  | 7.90676e+05  | 2.32404e+05  | 1.85240e+06  | 1.58135e+06  | 4.64807e+05  | 2.47954e+06 |
| 290   | 9.01281e+05  | 7.15517e+05  | 2.03862e+05  | 9.01288e+05  | 7.15523e+05  | 2.03863e+05  | 1.80257e+06  | 1.43104e+06  | 4.07725e+05  | 2.33739e+06 |
| 291   | 8.70186e+05  | 6.41699e+05  | 1.77437e+05  | 8.70193e+05  | 6.41704e+05  | 1.77438e+05  | 1.74038e+06  | 1.28340e+06  | 3.54875e+05  | 2.19134e+06 |
| 292   | 8.33109e+05  | 5.69732e+05  | 1.53162e+05  | 8.33115e+05  | 5.69737e+05  | 1.53163e+05  | 1.66622e+06  | 1.13947e+06  | 3.06325e+05  | 2.04170e+06 |
| 293   | 7.90282e+05  | 5.00119e+05  | 1.31039e+05  | 7.90288e+05  | 5.00123e+05  | 1.31040e+05  | 1.58057e+06  | 1.00024e+06  | 2.62080e+05  | 1.88875e+06 |
| 294   | 7.41975e+05  | 4.33341e+05  | 1.11041e+05  | 7.41981e+05  | 4.33344e+05  | 1.11042e+05  | 1.48396e+06  | 8.66685e+05  | 2.22084e+05  | 1.73280e+06 |
| 295   | 6.88493e+05  | 3.69858e+05  | 9.31114e+04  | 6.88499e+05  | 3.69861e+05  | 9.31122e+04  | 1.37699e+06  | 7.39719e+05  | 1.86224e+05  | 1.57416e+06 |
| 296   | 6.30171e+05  | 3.10101e+05  | 7.71637e+04  | 6.30176e+05  | 3.10104e+05  | 7.71644e+04  | 1.26035e+06  | 6.20205e+05  | 1.54328e+05  | 1.41313e+06 |
| 297   | 5.67373e+05  | 2.54472e+05  | 6.30859e+04  | 5.67378e+05  | 2.54474e+05  | 6.30865e+04  | 1.13475e+06  | 5.08946e+05  | 1.26172e+05  | 1.25004e+06 |
| 298   | 5.00491e+05  | 2.03338e+05  | 5.07402e+04  | 5.00497e+05  | 2.03340e+05  | 5.07408e+04  | 1.00099e+06  | 4.06678e+05  | 1.01481e+05  | 1.08520e+06 |
| 299   | 4.29941e+05  | 1.57029e+05  | 3.99653e+04  | 4.29947e+05  | 1.57031e+05  | 3.99659e+04  | 8.59888e+05  | 3.14061e+05  | 7.99312e+04  | 9.18929e+05 |
| 300   | 3.56158e+05  | 1.15838e+05  | 3.05787e+04  | 3.56165e+05  | 1.15840e+05  | 3.05792e+04  | 7.12323e+05  | 2.31679e+05  | 6.11579e+04  | 7.51544e+05 |
| 301   | 2.79597e+05  | 8.00145e+04  | 2.23788e+04  | 2.79603e+05  | 8.00164e+04  | 2.23793e+04  | 5.59200e+05  | 1.60031e+05  | 4.47581e+04  | 5.83367e+05 |
| 302   | 2.00724e+05  | 4.97645e+04  | 1.51477e+04  | 2.00731e+05  | 4.97663e+04  | 1.51483e+04  | 4.01455e+05  | 9.95308e+04  | 3.02960e+04  | 4.14717e+05 |
| 303   | 1.20020e+05  | 2.52500e+04  | 8.65409e+03  | 1.20028e+05  | 2.52516e+04  | 8.65464e+03  | 2.40048e+05  | 5.05016e+04  | 1.73087e+04  | 2.45913e+05 |
| 304   | 3.79724e+04  | 6.58639e+03  | 2.65566e+03  | 3.79808e+04  | 6.58785e+03  | 2.65625e+03  | 7.59531e+04  | 1.31742e+04  | 5.31191e+03  | 7.72700e+04 |
| 305   | -4.49275e+04 | -6.15743e+03 | -3.09749e+03 | -4.49183e+04 | -6.15617e+03 | -3.09686e+03 | -8.98458e+04 | -1.23136e+04 | -6.19436e+03 | 9.08970e+04 |
| 306   | -1.28186e+05 | -1.29598e+04 | -8.86004e+03 | -1.28176e+05 | -1.29588e+04 | -8.85934e+03 | -2.56362e+05 | -2.59186e+04 | -1.77194e+04 | 2.58277e+05 |
| 307   | -2.11312e+05 | -1.38461e+04 | -1.48882e+04 | -2.11301e+05 | -1.38454e+04 | -1.48875e+04 | -4.22613e+05 | -2.76914e+04 | -2.97757e+04 | 4.24564e+05 |
| 308   | -2.93819e+05 | -8.88815e+03 | -2.14369e+04 | -2.93807e+05 | -8.88779e+03 | -2.14360e+04 | -5.87625e+05 | -1.77759e+04 | -4.28729e+04 | 5.89455e+05 |
| 309   | -3.75228e+05 | 1.79640e+03  | -2.87563e+04 | -3.75215e+05 | 1.79634e+03  | -2.87553e+04 | -7.50443e+05 | 3.59274e+03  | -5.75116e+04 | 7.52652e+05 |

| phase | F+ x         | F+ y        | F+ z         | F- x         | F- y        | F- z         | Fnet x       | Fnet y      | Fnet z       | Fnet        |
|-------|--------------|-------------|--------------|--------------|-------------|--------------|--------------|-------------|--------------|-------------|
| 310   | -4.55074e+05 | 1.80456e+04 | -3.70890e+04 | -4.55060e+05 | 1.80450e+04 | -3.70879e+04 | -9.10134e+05 | 3.60907e+04 | -7.41769e+04 | 9.13865e+05 |
| 311   | -5.32904e+05 | 3.96547e+04 | -4.66674e+04 | -5.32888e+05 | 3.96536e+04 | -4.66661e+04 | -1.06579e+06 | 7.93083e+04 | -9.33335e+04 | 1.07281e+06 |
| 312   | -6.08281e+05 | 6.63781e+04 | -5.77103e+04 | -6.08285e+05 | 6.63763e+04 | -5.77087e+04 | -1.21655e+06 | 1.32754e+05 | -1.15419e+05 | 1.22920e+06 |
| 313   | -6.80792e+05 | 9.79318e+04 | -7.04207e+04 | -6.80774e+05 | 9.79293e+04 | -7.04188e+04 | -1.36157e+06 | 1.95861e+05 | -1.40839e+05 | 1.38277e+06 |
| 314   | -7.50041e+05 | 1.33996e+05 | -8.49831e+04 | -7.50022e+05 | 1.33993e+05 | -8.49809e+04 | -1.50006e+06 | 2.67988e+05 | -1.69964e+05 | 1.53326e+06 |
| 315   | -8.15659e+05 | 1.74218e+05 | -1.01561e+05 | -8.15639e+05 | 1.74213e+05 | -1.01559e+05 | -1.63130e+06 | 3.48431e+05 | -2.03120e+05 | 1.68042e+06 |
| 316   | -8.77304e+05 | 2.18214e+05 | -1.20297e+05 | -8.77283e+05 | 2.18209e+05 | -1.20294e+05 | -1.75459e+06 | 4.36423e+05 | -2.40590e+05 | 1.82399e+06 |
| 317   | -9.34662e+05 | 2.65577e+05 | -1.41305e+05 | -9.34640e+05 | 2.65571e+05 | -1.41302e+05 | -1.86930e+06 | 5.31148e+05 | -2.82607e+05 | 1.96374e+06 |
| 318   | -9.87446e+05 | 3.15875e+05 | -1.64677e+05 | -9.87424e+05 | 3.15868e+05 | -1.64674e+05 | -1.97487e+06 | 6.31742e+05 | -3.29351e+05 | 2.09945e+06 |
| 319   | -1.03541e+06 | 3.68657e+05 | -1.90477e+05 | -1.03538e+06 | 3.68648e+05 | -1.90472e+05 | -2.07079e+06 | 7.37305e+05 | -3.80949e+05 | 2.23090e+06 |
| 320   | -1.07832e+06 | 4.23457e+05 | -2.18737e+05 | -1.07829e+06 | 4.23447e+05 | -2.18732e+05 | -2.15661e+06 | 8.46904e+05 | -4.37469e+05 | 2.35788e+06 |
| 321   | -1.11600e+06 | 4.79799e+05 | -2.49464e+05 | -1.11597e+06 | 4.79788e+05 | -2.49458e+05 | -2.23197e+06 | 9.59587e+05 | -4.98922e+05 | 2.48021e+06 |
| 322   | -1.14829e+06 | 5.37199e+05 | -2.82634e+05 | -1.14827e+06 | 5.37187e+05 | -2.82627e+05 | -2.29656e+06 | 1.07439e+06 | -5.65261e+05 | 2.59770e+06 |
| 323   | -1.17508e+06 | 5.95171e+05 | -3.18192e+05 | -1.17506e+06 | 5.95158e+05 | -3.18185e+05 | -2.35014e+06 | 1.19033e+06 | -6.36377e+05 | 2.71017e+06 |
| 324   | -1.19629e+06 | 6.53230e+05 | -3.56057e+05 | -1.19626e+06 | 6.53216e+05 | -3.56049e+05 | -2.39255e+06 | 1.30645e+06 | -7.12106e+05 | 2.81748e+06 |
| 325   | -1.21186e+06 | 7.10894e+05 | -3.96116e+05 | -1.21183e+06 | 7.10879e+05 | -3.96108e+05 | -2.42369e+06 | 1.42177e+06 | -7.92224e+05 | 2.91947e+06 |
| 326   | -1.22178e+06 | 7.67693e+05 | -4.38230e+05 | -1.22175e+06 | 7.67677e+05 | -4.38221e+05 | -2.44352e+06 | 1.53537e+06 | -8.76451e+05 | 3.01601e+06 |
| 327   | -1.22607e+06 | 8.23168e+05 | -4.82232e+05 | -1.22604e+06 | 8.23151e+05 | -4.82222e+05 | -2.45211e+06 | 1.64632e+06 | -9.64454e+05 | 3.10699e+06 |
| 328   | -1.22478e+06 | 8.76877e+05 | -5.27930e+05 | -1.22476e+06 | 8.76859e+05 | -5.27919e+05 | -2.44954e+06 | 1.75374e+06 | -1.0558e+06  | 3.19228e+06 |
| 329   | -1.21802e+06 | 9.28398e+05 | -5.75108e+05 | -1.21799e+06 | 9.28379e+05 | -5.75097e+05 | -2.43601e+06 | 1.85678e+06 | -1.15021e+06 | 3.27181e+06 |
| 330   | -1.20588e+06 | 9.77331e+05 | -6.23529e+05 | -1.20585e+06 | 9.77311e+05 | -6.23517e+05 | -2.41173e+06 | 1.95464e+06 | -1.24705e+06 | 3.34547e+06 |
| 331   | -1.18852e+06 | 1.02330e+06 | -6.72935e+05 | -1.18849e+06 | 1.02328e+06 | -6.72922e+05 | -2.37701e+06 | 2.04659e+06 | -1.34586e+06 | 3.41322e+06 |
| 332   | -1.16612e+06 | 1.06597e+06 | -7.23050e+05 | -1.16609e+06 | 1.06595e+06 | -7.23036e+05 | -2.33221e+06 | 2.13192e+06 | -1.44609e+06 | 3.47498e+06 |
| 333   | -1.13887e+06 | 1.10502e+06 | -7.73585e+05 | -1.13885e+06 | 1.10500e+06 | -7.73570e+05 | -2.27773e+06 | 2.21002e+06 | -1.54716e+06 | 3.53071e+06 |
| 334   | -1.10702e+06 | 1.14018e+06 | -8.24237e+05 | -1.10700e+06 | 1.14015e+06 | -8.24222e+05 | -2.21402e+06 | 2.28033e+06 | -1.64846e+06 | 3.58040e+06 |
| 335   | -1.07081e+06 | 1.17119e+06 | -8.74694e+05 | -1.07079e+06 | 1.17117e+06 | -8.74678e+05 | -2.14161e+06 | 2.34235e+06 | -1.74937e+06 | 3.62401e+06 |
| 336   | -1.03052e+06 | 1.19785e+06 | -9.24637e+05 | -1.03050e+06 | 1.19783e+06 | -9.24621e+05 | -2.06103e+06 | 2.39568e+06 | -1.84926e+06 | 3.66154e+06 |
| 337   | -9.86443e+05 | 1.21999e+06 | -9.73744e+05 | -9.86426e+05 | 1.21997e+06 | -9.73727e+05 | -1.97287e+06 | 2.43997e+06 | -1.94747e+06 | 3.69301e+06 |
| 338   | -9.38883e+05 | 1.23749e+06 | -1.02169e+06 | -9.38868e+05 | 1.23746e+06 | -1.02167e+06 | -1.87775e+06 | 2.47495e+06 | -2.04336e+06 | 3.71842e+06 |
| 339   | -8.88170e+05 | 1.25023e+06 | -1.06816e+06 | -8.88156e+05 | 1.25021e+06 | -1.06814e+06 | -1.77633e+06 | 2.50045e+06 | -2.13630e+06 | 3.73783e+06 |
| 340   | -8.34641e+05 | 1.25819e+06 | -1.11282e+06 | -8.34628e+05 | 1.25817e+06 | -1.11281e+06 | -1.66927e+06 | 2.51636e+06 | -2.22563e+06 | 3.75126e+06 |
| 341   | -7.78644e+05 | 1.26133e+06 | -1.15539e+06 | -7.78632e+05 | 1.26131e+06 | -1.15537e+06 | -1.55728e+06 | 2.52264e+06 | -2.31075e+06 | 3.75878e+06 |
| 342   | -7.20533e+05 | 1.25969e+06 | -1.19554e+06 | -7.20523e+05 | 1.25967e+06 | -1.19553e+06 | -1.44106e+06 | 2.51935e+06 | -2.39107e+06 | 3.76045e+06 |
| 343   | -6.60670e+05 | 1.25332e+06 | -1.23301e+06 | -6.60661e+05 | 1.25330e+06 | -1.23300e+06 | -1.32133e+06 | 2.50661e+06 | -2.46601e+06 | 3.75636e+06 |
| 344   | -5.99418e+05 | 1.24232e+06 | -1.26753e+06 | -5.99410e+05 | 1.24230e+06 | -1.26751e+06 | -1.19883e+06 | 2.48462e+06 | -2.53504e+06 | 3.74659e+06 |
| 345   | -5.37142e+05 | 1.22682e+06 | -1.29884e+06 | -5.37135e+05 | 1.22681e+06 | -1.29882e+06 | -1.07428e+06 | 2.45363e+06 | -2.59766e+06 | 3.73125e+06 |
| 346   | -4.74204e+05 | 1.20700e+06 | -1.32672e+06 | -4.74199e+05 | 1.20698e+06 | -1.32670e+06 | -9.48403e+05 | 2.41398e+06 | -2.65342e+06 | 3.71044e+06 |
| 347   | -4.10964e+05 | 1.18303e+06 | -1.35096e+06 | -4.10959e+05 | 1.18302e+06 | -1.35094e+06 | -8.21924e+05 | 2.36606e+06 | -2.70189e+06 | 3.68429e+06 |
| 348   | -3.47774e+05 | 1.15516e+06 | -1.37137e+06 | -3.47770e+05 | 1.15515e+06 | -1.37136e+06 | -6.95545e+05 | 2.31031e+06 | -2.74273e+06 | 3.65293e+06 |
| 349   | -2.84980e+05 | 1.12363e+06 | -1.38782e+06 | -2.84977e+05 | 1.12361e+06 | -1.38780e+06 | -5.69957e+05 | 2.24724e+06 | -2.77562e+06 | 3.61649e+06 |
| 350   | -2.22916e+05 | 1.08870e+06 | -1.40016e+06 | -2.22914e+05 | 1.08869e+06 | -1.40014e+06 | -4.45830e+05 | 2.17739e+06 | -2.80030e+06 | 3.57512e+06 |
| 351   | -1.61907e+05 | 1.05068e+06 | -1.40830e+06 | -1.61906e+05 | 1.05067e+06 | -1.40829e+06 | -3.23813e+05 | 2.10136e+06 | -2.81658e+06 | 3.52898e+06 |
| 352   | -1.02263e+05 | 1.00988e+06 | -1.41217e+06 | -1.02262e+05 | 1.00987e+06 | -1.41216e+06 | -2.04525e+05 | 2.01975e+06 | -2.82433e+06 | 3.47823e+06 |
| 353   | -4.42790e+04 | 9.66621e+05 | -1.41117e+06 | -4.42786e+04 | 9.66613e+05 | -1.41172e+06 | -8.85577e+04 | 1.93323e+06 | -2.82346e+06 | 3.42304e+06 |
| 354   | 1.17660e+04  | 9.21241e+05 | -1.40699e+06 | 1.17659e+04  | 9.21234e+05 | -1.40697e+06 | 2.35319e+04  | 1.84247e+06 | -2.81396e+06 | 3.36358e+06 |

| phase | F+ x        | F+ y        | F+ z         | F- x        | F- y        | F- z         | Fnet x      | Fnet y      | Fnet z       | Fnet        |
|-------|-------------|-------------|--------------|-------------|-------------|--------------|-------------|-------------|--------------|-------------|
| 355   | 6.56105e+04 | 8.74085e+05 | -1.39794e+06 | 6.56100e+04 | 8.74079e+05 | -1.39793e+06 | 1.31220e+05 | 1.74816e+06 | -2.79587e+06 | 3.30003e+06 |
| 356   | 1.17012e+05 | 8.25505e+05 | -1.38466e+06 | 1.17011e+05 | 8.25499e+05 | -1.38465e+06 | 2.34023e+05 | 1.65100e+06 | -2.76930e+06 | 3.23259e+06 |
| 357   | 1.65747e+05 | 7.75852e+05 | -1.36721e+06 | 1.65746e+05 | 7.75847e+05 | -1.36720e+06 | 3.31493e+05 | 1.55170e+06 | -2.73442e+06 | 3.16144e+06 |
| 358   | 2.11614e+05 | 7.25478e+05 | -1.34572e+06 | 2.11613e+05 | 7.25473e+05 | -1.34571e+06 | 4.23227e+05 | 1.45095e+06 | -2.69144e+06 | 3.08678e+06 |
| 359   | 2.54432e+05 | 6.74730e+05 | -1.32032e+06 | 2.54430e+05 | 6.74726e+05 | -1.32032e+06 | 5.08862e+05 | 1.34946e+06 | -2.64064e+06 | 3.00881e+06 |
| 360   | 2.94040e+05 | 6.23948e+05 | -1.29118e+06 | 2.94038e+05 | 6.23944e+05 | -1.29117e+06 | 5.88079e+05 | 1.24789e+06 | -2.58235e+06 | 2.92773e+06 |

## Appendix D. Formula and variable ledger

| Symbol          | Meaning                                                | Units or type                         |
|-----------------|--------------------------------------------------------|---------------------------------------|
| Q               | Authoritative phase-transition and kinetic-cost metric | dimensionless 3 x 3 SPD matrix        |
| c_hat           | Earth-CMB unit direction and axis B                    | unit vector                           |
| V_E             | Earth-CMB velocity vector                              | m/s                                   |
| e0              | Bar transverse reference at phi=0                      | unit vector                           |
| A0              | Axis A reference at phi=0                              | unit vector                           |
| phi             | Outer phase, omega_B t                                 | radian                                |
| u(phi)          | Positive endpoint radial direction                     | unit vector                           |
| u_phi           | du/dphi                                                | dimensionless derivative              |
| R               | Mass radius from bar center                            | m                                     |
| m               | Each endpoint mass                                     | kg                                    |
| omega_B         | Outer angular speed                                    | rad/s                                 |
| omega_A         | Inner angular speed, 2 omega_B                         | rad/s                                 |
| kappa_F         | Constitutive Q-metric mechanical gain                  | dimensionless in native normalization |
| B(phi)          | $u_{\phi}^T Q V_E$                                     | m/s                                   |
| C(phi)          | $u_{\phi}^T Q u_{\phi}$                                | dimensionless                         |
| F_plus, F_minus | Individual Q-metric radial force vectors               | N                                     |
| F_net           | $F_{\text{plus}} + F_{\text{minus}}$                   | N                                     |
| J_B             | Integral of F_net over one outer period                | N s                                   |

## References and source ledger

1. K-Field Maximum Residual Closed CMB Route, source JSON epoch 1784650136. Supplies Q, Earth-CMB unit direction, optimized funded and return geometry, Cell3 basis, vertices, nodes, and face data.
2. Cell3 Earth-CMB Vector Derivation, epoch 1783520047. Supplies the exact-SI Earth-CMB velocity vector and magnitude.
3. K-field Rotor Blade Analogue, epoch 1783267951. Supplies the translating-rotor Q-metric expansion and radial force channel.
4. K-Field Transition-Cost Ellipsoid, epoch 1784515891. Supplies the negative-normal gravity and transition-cost geometry.
5. K-Field 2:1 force time series, one-degree numerical packet generated from the exact apparatus geometry and force law.

## Archival colophon

| Field                      | Entry                                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| Document title             | K-Field Dual-Axis Kinetic Mass Analysis                                                                         |
| Principal Investigator     | P. Dwayne Esterline                                                                                             |
| Organization               | Krytronx                                                                                                        |
| Output filename            | K-Field_dual_axis_kinetic_mass_analysis_1784741897.pdf                                                          |
| Machine-readable companion | K-Field_dual_axis_kinetic_mass_analysis_1784741897.json                                                         |
| Generation epoch           | 1784741897                                                                                                      |
| Page size                  | US Letter; landscape pages used for complete force table                                                        |
| Coordinate frame           | ICRS Cartesian                                                                                                  |
| Cell3 units                | Native source length units                                                                                      |
| Mechanical units           | SI; forces reported at $\omega_B=1$ rad/s and $\kappa_F=1$ unless otherwise stated                              |
| Fonts                      | Helvetica, Helvetica-Bold, Times-Roman, Courier                                                                 |
| Classification             | Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute. |